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Comparing Kappa Weighting and Causal Machine Learning Estimators via Weight-Based Diagnostics

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Abstract

This thesis applies the outcome-weights framework of **?** to kappa-based local average treatment effect (LATE) estimators of **?** and **?**. Any estimator that can be written as a weighted sum of observed outcomes, $\hat{\tau} = \sum_i \omega_i Y_i$, admits direct covariate balance diagnostics via Love plots and effective sample size (ESS) calculations. Knaus (2024) derives such outcome weights for double machine learning (DML) and generalised random forest estimators; Appendix A.4 sketches but does not implement the same for kappa estimators. We fill this gap by deriving closed-form outcome weights for all five kappa-based LATE estimators and showing that the sum-to-zero condition $\sum_i \omega_i = 0$ is the exact finite-sample criterion for translation invariance—the property that the LATE estimate is unaffected by additive rescaling of the outcome variable.

We apply the unified outcome-weight diagnostics to three empirical settings: the Vietnam draft lottery (**?**), college proximity as an instrument for education (**?**), and sibling sex composition as an instrument for fertility (**?**). A striking finding across applications is that in low-dimensional, near-random-assignment designs, ESS and the percentage of negative weights are virtually identical across all estimators and all machine learning learners—a pattern we term *design dominates learner*. Normalization, not covariate adjustment, drives estimate divergence. Unnormalized kappa estimators fail translation invariance dramatically in the Angrist–Evans design, with estimates changing sign under outcome recoding. Normalized estimators $\hat{\tau}_u$ and $\hat{\tau}_{a,10}$ are robust throughout.

Keywords: local average treatment effect, instrumental variables, kappa weighting, outcome weights, translation invariance, covariate balance, double machine learning.

JEL classification: C14, C21, C26.

Contents

List of Figures

List of Tables

1 Econometric Framework

1.1 Instrumental variables LATE, and compliers

The causal effect of schooling on later earnings is a classical problem in empirical economics. Individuals with more schooling often display higher wages, but this gap cannot be interpreted directly as the effect of education. Schooling decissions are are related to family background, ability, motivation, and local opportunities, which also affect later labour-market outcomes. A naive comparison between more and less educated individuals would therefore combine the causal effect of schooling with selection into education. Instrumental variables can be a way to adress this problem. The instrument has the purpose of isolating the variation in the treatment that is induced by an external source. This is done by shifting the probability of treatment while being otherwise unrelated to the potential outcomes, except through the treatment itself.
In the classical linear IV model, this logic is often introduced through two-stage least squares (2SLS). Consider the structural equation

$$Y_i = \alpha + \theta D_i + U_i, \tag{1}$$

where D_i is the treatment, Y_i is the outcome and U_i entails the unobserved determinants of the outcome. Ordinary least squares is generally inconsistent if D_i is endogenous, that is, if D_i is correlated with U_i. 2SLS adresses this problem by using only the part of the treatment variation that is predicted by the instrument. In the first stage, the endogenous treatment is regressed on the instrument.

$$D_i = \pi_0 + \pi_1 Z_i + v_i, \tag{2}$$

This regression captures the relationship between the instrument and treatment take-up. The fitted values from this regression are denoted by $\widehat{D}_i$. In the second stage, the outcome is regressed on this predicted treatment,

$$Y_i = \alpha + \theta \widehat{D}_i + \varepsilon_i. \tag{3}$$

The second stage hence uses the variation in treatment that is induced by the instrument.
For this procedure in order to have consistent estimators two important requirements must be fullfilled (Wooldridge, 2016):

$$\mathrm{Cov}(Z_i, D_i) \neq 0, \qquad \mathrm{Cov}(Z_i, U_i) = 0 \tag{4}$$

The first is relevance, which means that the instrument affects treatment take up. The second is exogeneity, which means that the the instrument is unrelated to the unobserved determinants of the outcome. Under homogeneous treatment effects, 2SLS recovers the common causal effect θ. For a binary instrument and a binary treatment, the corresponding IV estimand can be written as the Wald ratio (Wald, 1940):

$$\theta^{Wald} = \frac{E[Y|Z=1] - E[Y|Z=0]}{E[D|Z=1] - E[D|Z=0]} \tag{5}$$

The numerator is the reduced-form effect of the instrument on the outcome. It measures how average outcomes differ between individuals with $Z_i = 1$ and individuals with $Z_i = 0$. The denominator is the first stage. It measures how strongly the instrument shifts the probability of receiving treatment.Hence, the Wald ratio scales the effect of the instrument on the outcome by the effect of the instrument on treatment.

In the just-identified case with a binary instrument and no additional covariates, the 2SLS coefficient is algebraically equivalent to this Wald ratio. Under homogeneous treatment effects, this implies $\theta^{Wald} = \theta$.

Once treatment effects are allowed to differ across individuals, the Wald ratio no longer generally represents the average treatment effect in the full population. The reason is that the instrument does not necessarily change treatment status for every individual. Instead, the reduced form is generated by those individuals whose treatment status reacts to the instrument. This motivates the potential-outcomes formulation of instrumental variables. Let $D_i(1)$ and $D_i(0)$ denote individual (i)'s potential treatment status under instrument values $Z_i = 1$ and $Z_i = 0$, respectively. The observed treatment status is therefore

$$D_i = Z_i D_i(1) + (1 - Z_i) D_i(0). \tag{6}$$

This notation makes explicit that individuals may differ in how their treatment status responds to the instrument. Some individuals may take the treatment regardless of the instrument, some may never take it, and others may be induced into treatment only when $(Z_i = 1)$. Before imposing the exclusion restriction, potential outcomes may depend on both the instrument and the treatment. I denote these potential outcomes by $Y_i(z, d)$, where $z \in \{0, 1\}$ is the instrument value and $d \in \{0, 1\}$ is the treatment status. Thus, $Y_i(1, 0)$ is the outcome individual i would realize if assigned $Z_i = 1$ but not treated, while $Y_i(1, 1)$ is the outcome under $Z_i = 1$ and treatment.

The exclusion restriction rules out a direct effect of the instrument on the outcome. Formally,

$$Y_i(1, d) = Y_i(0, d) = Y_i(d), \qquad d \in \{0, 1\} \tag{7}$$

After imposing this restriction, the instrument can affect the outcome only through treatment. The potential outcomes can therefore be written simply as $Y_i(1)$ and $Y_i(0)$, and the observed outcome satisfies

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0) \tag{8}$$

The pair $(D_i(1), D_i(0))$ classifies individuals into four compliance types (Angrist et al., 1996):

	$D_i(0) = 0$	$D_i(0) = 1$
$D_i(1) = 0$	Never-taker	Defier
$D_i(1) = 1$	Complier	Always-taker

Never-takers are untreated irrespective of the instrument, $D_i(1) = D_i(0) = 0$. Always-takers are treated irrespective of the instrument, $D_i(1) = D_i(0) = 1$. Compliers are induced into treatment by the instrument, $D_i(1) = 1$ and $D_i(0) = 0$. Defiers move in the opposite direction, $D_i(1) = 0$ and $D_i(0) = 1$. The assumption of monotonicity rules out the existence of defiers, requiring that $D_i(1) \geq D_i(0)$ for all i. This classification is important because the first stage is determined by the change in treatment status induced by the instrument:

$$D_i(1) - D_i(0). \tag{9}$$

For never-takers and always-takers this difference is zero, so these groups do not contribute to the first stage. For compliers it equals one, while for defiers it equals minus one. Without further restrictions, the reduced form may therefore combine treatment effects of compliers and defiers with opposite signs. In order to rule out defiers the monotonicity assumption is imposed

$$D_i(1) \geq D_i(0) \tag{10}$$

for all (i). Hence, the instrument may increase treatment take-up, but it may not induce any individual to move in the opposite direction. Hence, the denominator of the Wald ratio measures the size of the complier group. Thus, the same Wald ratio has two different interpretations depending on the maintained assumptions. Under homogeneous treatment effects, it recovers the common treatment effect. Under heterogeneous treatment effects, instrument independence, exclusion, a non-zero first stage, and monotonicity, it identifies the Local Average Treatment Effect, meaning the average treatment effect for compliers.

1.1.1 The importance of weights

The previous argument establishes the LATE interpretation in the binary-instrument, binary-treatment setting. This benchmark is most transparent when the instrument comparison is valid unconditionally. In many empirical applications, however, instrument validity is more plausibly stated conditional on observed covariates. Individuals differ in observed characteristics (X_i), and these characteristics may be related to instrument assignment. For example, in a schooling application, college proximity may be related to region, family background, or urban residence. A simple comparison of individuals with $Z_i = 1$ and $Z_i = 0$ would then not compare otherwise similar individuals.

The IV independence assumption is therefore more naturally formulated conditionally,

$$Y_i(0,0), Y_i(1,0), Y_i(0,1), Y_i(1,1), D_i(0), D_i(1) \perp Z_i \mid X_i. \tag{11}$$

Under this condition, the relevant comparison is no longer the unconditional contrast between all individuals with $Z_i = 1$ and all individuals with $Z_i = 0$, but a comparison within covariate strata. This introduces the conditional instrument propensity score

$$p(X_i) = P(Z_i = 1 \mid X_i), \tag{12}$$

which measures the probability of receiving the instrument for individuals with observed characteristics X_i.

By Frisch–Waugh–Lovell, Z_i can be replaced by its residual after projecting it on X_i,

$$\tilde{Z}_i = Z_i - L[Z \mid X_i]. \tag{13}$$

This removes the part that is explained by the covariates from the instrument, and uses only the remaining instrument variation. But Blandhol et al. (2022) show that this only gives a clean LATE-type interpretation if the linear projection correctly reproduces the true conditional mean of the instrument,

$$\mathbb{L}[Z \mid X] = \mathbb{E}[Z \mid X] = \mathbb{P}(Z = 1 \mid X), \tag{14}$$

a condition also referred to as the rich covariate condition. Saturated covariate specifications guarantee it, but additive controls typically do not. If the condition fails, covariate-adjusted TSLS may no longer average only complier effects, but can also depend on always-takers or never-takers.

This motivates an alternative route that uses the conditional instrument propensity score explicitly, rather than through a linear projection of Z_i on X_i. The kappa theorem in Abadie (2003) provides such a route, identifying moments of the complier population as reweighted moments of the observed data, under the conditions stated next.

1.2 Abadie's kappa weights

The following four conditions, due to Abadie (2003), extend the requirements introduced in Section 2.1 to settings with covariates.

Assumption (Conditional IV).

(i) Independence: $Y_i(0,0), Y_i(1,0), Y_i(0,1), Y_i(1,1), D_i(0), D_i(1) \perp Z_i \mid X_i$.

(ii) Exclusion: $Y_i(1,d) = Y_i(0,d)$ almost surely conditional on X_i, for $d \in \{0,1\}$.

(iii) First stage and overlap: $0 < p(X_i) < 1$, and $P(D_i(1) = 1 \mid X_i) > P(D_i(0) = 1 \mid X_i)$.

(iv) Monotonicity: $P(D_i(1) \geq D_i(0) \mid X_i) = 1$.

Condition (i) restates the conditional independence assumption introduced above. Conditions (ii) and (iv) are the conditional analogues of the exclusion restriction and the monotonicity assumption from Section 2.1. Condition (iii) adds an overlap requirement, ruling out values of X_i for which one of the two instrument values is never observed. Define

$$\kappa_{0i} = (1 - D_i)\,\frac{(1 - Z_i) - (1 - p(X_i))}{p(X_i)\big(1 - p(X_i)\big)}, \qquad \kappa_{1i} = D_i\,\frac{Z_i - p(X_i)}{p(X_i)\big(1 - p(X_i)\big)}, \tag{15}$$

$$\kappa_i = \kappa_{0i}\big(1 - p(X_i)\big) + \kappa_{1i}\,p(X_i) = 1 - \frac{D_i(1 - Z_i)}{1 - p(X_i)} - \frac{(1 - D_i)Z_i}{p(X_i)}. \tag{16}$$

Lemma (Abadie's kappa theorem; Abadie, 2003). Let $g(\cdot)$, $g_0(\cdot)$, and $g_1(\cdot)$ be measurable functions with finite first moments. Under Assumption (Conditional IV),

$$E\Big[g(Y_i, D_i, X_i) \mid D_i(1) > D_i(0)\Big] = \frac{E\Big[\kappa_i\, g(Y_i, D_i, X_i)\Big]}{P(D_i(1) > D_i(0))}, \tag{a}$$

$$E\Big[g_0(Y_i(0), X_i) \mid D_i(1) > D_i(0)\Big] = \frac{E\Big[\kappa_{0i}\, g_0(Y_i, X_i)\Big]}{P(D_i(1) > D_i(0))}, \tag{b}$$

$$E\Big[g_1(Y_i(1), X_i) \mid D_i(1) > D_i(0)\Big] = \frac{E\Big[\kappa_{1i}\, g_1(Y_i, X_i)\Big]}{P(D_i(1) > D_i(0))}. \tag{c}$$

Equations (a)-(c) also hold conditional on X_i. This lemma is what makes the complier subpopulation statistically tractable despite the fact that no individual can ever be classified as a complier: $D_i(0)$ and $D_i(1)$ are never jointly observed, so a sample of compliers cannot be constructed directly. Instead, any moment of interest for compliers can be recovered as a reweighted moment of the full observed sample, where the weight depends only on observable quantities (D_i, Z_i, X_i).

Two applications of this lemma are needed to identify τ_{LATE}. *The denominator.* Setting $g(Y_i, D_i, X_i) = 1$ in (a) gives

$$E[\kappa_i] = P(D_i(1) > D_i(0)). \tag{17}$$

The complier share is therefore identified as the average of κ_i over the full sample.

The numerator. Setting $g_0(Y_i(0), X_i) = Y_i(0)$ in (b) and $g_1(Y_i(1), X_i) = Y_i(1)$ in (c) gives

$$E[Y_i(0) \mid D_i(1) > D_i(0)] = \frac{E[\kappa_{0i} Y_i]}{P(D_i(1) > D_i(0))}, \qquad E[Y_i(1) \mid D_i(1) > D_i(0)] = \frac{E[\kappa_{1i} Y_i]}{P(D_i(1) > D_i(0))}. \tag{18}$$

Subtracting the two and using the definition of τ_{LATE},

$$\tau_{LATE} = \frac{E[\kappa_{1i} Y_i] - E[\kappa_{0i} Y_i]}{P(D_i(1) > D_i(0))} = \frac{E\big[(\kappa_{1i} - \kappa_{0i}) Y_i\big]}{P(D_i(1) > D_i(0))}. \tag{19}$$

The weight difference simplifies, since

$$\kappa_{1i} - \kappa_{0i} = D_i \frac{Z_i - p(X_i)}{p(X_i)(1 - p(X_i))} - (1 - D_i) \frac{p(X_i) - Z_i}{p(X_i)(1 - p(X_i))} = \frac{Z_i - p(X_i)}{p(X_i)(1 - p(X_i))}, \tag{20}$$

which no longer depends on D_i. Hence

$$\tau_{LATE} = \frac{1}{P(D_i(1) > D_i(0))} E\left[Y_i \frac{Z_i - p(X_i)}{p(X_i)\big(1 - p(X_i)\big)} \right]. \tag{21}$$

It is worth noting already that $P(D_i(1) > D_i(0))$ can equivalently be written as $E[\kappa_i]$, $E[\kappa_{0i}]$, or $E[\kappa_{1i}]$, all three are valid in population under Assumption (Conditional IV), they coincide in population, but their sample analogues need not coincide in finite samples whether $p(X_i)$ is known or estimated. This ambiguity in the denominator is the source of the different weighting estimators of τ_{LATE} developed in the next section.

1.3 Translation invariance

The outcome-weight representation makes it possible to distinguish several finite-sample properties that are closely related but conceptually different. Throughout this section, consider an estimator that admits the representation

$$\hat{\tau} = \sum_{i=1}^{N} \omega_i Y_i = \omega' Y, \tag{22}$$

where the weights ω_i are the final outcome weights of the estimator. These weights should be distinguished from identification weights such as Abadie's κ-weights. The latter identify moments of the complier population, whereas the former describe how the finite-sample estimator combines the observed outcomes.
Translation invariance requires that adding a constant to all outcomes does not change the treatment effect estimate. Formally, for any constant c,

$$\hat{\tau}(Y + c\mathbf{1}_N) = \hat{\tau}(Y). \tag{23}$$

Using the outcome-weight representation in (22), the effect of such a shift is

$$\begin{aligned} \hat{\tau}(Y + c\mathbf{1}_N) &= \sum_{i=1}^{N} \omega_i (Y_i + c) \\ &= \sum_{i=1}^{N} \omega_i Y_i + c \sum_{i=1}^{N} \omega_i \\ &= \hat{\tau}(Y) + c \sum_{i=1}^{N} \omega_i. \end{aligned} \tag{24}$$

Hence,

$$\hat{\tau}(Y + c\mathbf{1}_N) = \hat{\tau}(Y) \quad \Longleftrightarrow \quad \sum_{i=1}^{N} \omega_i = 0. \tag{25}$$

The total sum of the outcome weights is therefore an exact finite-sample criterion for translation invariance. If $\sum_i \omega_i \neq 0$, the estimator changes by $c \sum_i \omega_i$ under an additive shift of size c. Thus, $\sum_i \omega_i$ measures the per-unit sensitivity of the estimator to additive recodings of the outcome. This property is particularly relevant for log outcomes. If wages are measured in cents instead of dollars before taking logs, then

$$\log(100Y_i) = \log(Y_i) + \log(100). \tag{26}$$

Changing the unit of the untransformed wage variable therefore corresponds to an additive shift of the log outcome. A translation-invariant estimator is unaffected by this transformation, whereas an estimator with $\sum_i \omega_i \neq 0$ changes by $\log(100) \sum_i \omega_i$.

Scale equivariance refers to the behavior of the estimator under transformations of the scale of the outcome variable. In the context of log outcomes, the most important implication is invariance to multiplicative changes in the untransformed outcome. Since multiplying the original outcome by a positive constant $a > 0$ changes the log outcome by an additive constant,

$$\log(aY_i) = \log(Y_i) + \log(a), \tag{27}$$

translation invariance of the estimator in log outcomes implies invariance to the units in which the original outcome is measured. Thus, for log outcomes, scale invariance with respect to the original variable is closely connected to translation invariance of the transformed outcome. It is important, however, not to confuse scale equivariance with normalization of the outcome weights. Scale equivariance concerns transformations of the outcome variable. Normalization concerns the total mass of the final outcome weights assigned to treated and untreated observations.

A stronger property than translation invariance is full normalization of the outcome weights. In treatment-effect settings with binary treatment D_i, the weights are fully normalized if

$$\sum_{i=1}^{N} \omega_i D_i = 1 \quad \text{and} \quad \sum_{i=1}^{N} \omega_i (1 - D_i) = -1. \tag{28}$$

The first condition says that the total weight assigned to treated observations equals one. The second says that the total weight assigned to untreated observations equals minus one. Under these two conditions,

$$\sum_{i=1}^{N} \omega_i = \sum_{i=1}^{N} \omega_i D_i + \sum_{i=1}^{N} \omega_i (1 - D_i) = 1 - 1 = 0. \tag{29}$$

Full normalization therefore implies translation invariance. The converse does not generally hold.

The interpretation of full normalization is straightforward. If (28) holds, the estimator can be viewed as a normalized weighted treated mean minus a normalized weighted untreated mean:

$$\hat{\tau} = \sum_{D_i=1} \omega_i Y_i + \sum_{D_i=0} \omega_i Y_i, \qquad \sum_{D_i=1} \omega_i = 1, \quad \sum_{D_i=0} \omega_i = -1. \tag{30}$$

Thus, full normalization gives the outcome-weight representation the intuitive structure of a treated-control contrast.

Partial normalization and the relation to translation invariance. Some estimators normalize only one side of the comparison. For example, an estimator may satisfy

$$\sum_{i=1}^{N} \omega_i D_i = 1, \qquad \sum_{i=1}^{N} \omega_i (1 - D_i) \neq -1. \tag{31}$$

In this case, the treated part of the estimator has the interpretation of a proper weighted treated mean, but the untreated part does not have the corresponding normalized control-mean interpretation. Moreover,

$$\sum_{i=1}^{N} \omega_i = 1 + \sum_{i=1}^{N} \omega_i (1 - D_i), \tag{32}$$

so translation invariance generally fails unless the untreated weights happen to sum to -1. Analogously, if the untreated weights sum to -1 but the treated weights do not sum to 1, translation invariance also generally fails.

A different case arises when the total weight sum is zero but the treated and untreated components are not separately normalized. For example, it may be that

$$\sum_{i=1}^{N} \omega_i D_i = a, \qquad \sum_{i=1}^{N} \omega_i (1 - D_i) = -a, \qquad a \neq 1. \tag{33}$$

Then $\sum_i \omega_i = 0$, so the estimator is translation invariant. However, the treated and untreated components do not separately correspond to normalized weighted means. Such an estimator is stable to additive recodings of the outcome, but it lacks the stronger interpretation of a fully normalized treated-control contrast.

Table 1 summarizes the distinction between these properties.

Table 1: Outcome-weight properties and their interpretation

Property	Algebraic condition	Interpretation
Translation invariance	$\sum_i \omega_i = 0$	The estimator is invariant to additive shifts of the outcome, $Y_i \mapsto Y_i + c$.
Treated-side normalization	$\sum_i \omega_i D_i = 1$	The treated component has the scale of a normalized weighted treated mean.
Untreated-side normalization	$\sum_i \omega_i (1 - D_i) = -1$	The untreated component has the scale of minus a normalized weighted untreated mean.
Full normalization	$\sum_i \omega_i D_i = 1$ and $\sum_i \omega_i (1 - D_i) = -1$	The estimator can be interpreted as a normalized weighted treated mean minus a normalized weighted untreated mean. This implies translation invariance.
Translation invariance without full normalization	$\sum_i \omega_i = 0$, but treated and untreated sums are not 1 and -1	The estimator is stable to additive outcome shifts but does not have the full treated-control weighted-mean interpretation.
Scale invariance in log outcomes	$\hat{\tau}(\log(aY)) = \hat{\tau}(\log Y)$	The estimate in logs is invariant to the unit of measurement of the original outcome. This follows from translation invariance in log outcomes.

1.4 Cross Fitting

The DML estimators used in this thesis rely on nuisance functions that are estimated before the final treatment-effect parameter is computed. The reason for this seperation is that flexible machine-learning methods can overfit. If the same observations are used both to train the nuisance functions and to evaluate the estimating equation, the nuisance estimation error can become mechanically related to the score residuals. This can create finite-sample bias even when the nuisance learner has good predictive performance. Double Machine Learning addresses this problem with cross fitting (Chernozhukov et al., 2018). For this K-fold cross fitting procedure is used in which the sample is first split into K folds. For each fold, the nuisance functions are estimated using only the observations outside that fold. Formally, let $I_1, \ldots, I_K$ be a partition of the sample indices $\{1, \ldots, N\}$. For each fold I_k, the nuisance vector is estimated on the complement $I_k^c = \{1, \ldots, N\} \setminus I_k$. For an observation $i \in I_k$, the cross-fitted nuisance prediction is therefore given by

$$\hat{\eta}_i = \hat{\eta}_{-k}(X_i),$$

where $\hat{\eta}_{-k}$ denotes the nuisance functions trained without using observations in fold I_k. These fitted nuisance functions are then used to predict the nuisance components for the held-out observations. Repeating this procedure over all folds produces out-of-sample nuisance predictions for every observation. The final DML estimate is then obtained from the orthogonal score using these cross-fitted predictions. It is important to mention that each observation is used twice, but for different purposes. It is used to train nuisance functions for other folds, and it is used in the final score only with nuisance predictions obtained from models that were not trained on that observation. This preserves the practical advantage of using the full sample while avoiding the direct own-observation overfitting problem. Cross-fitting is combined with Neyman orthogonal scores. Orthogonality means that small first-stage errors in the nuisance functions have only a second-order effect on the final parameter estimate. Intuitively, the score is constructed such that the treatment-effect estimate is locally insensitive to small perturbations in the nuisance estimates. This makes it possible to use tree based learners such as random forest or xgboost without treating the resulting causal estimate as a simple plug-in machine-learning prediction. In the empirical applications below, I use five-fold cross-fitting. The choice of five folds is due to the reason that moderate fold numbers keep enough observations for nuisance training while still ensuring that the score is evaluated out of sample.

1.4.1 Nuisance Training

The cross fitting procedure determines on which observations the nuisance functions are trained and evaluated. However it does not determine how complex a nuisance learner should be. This is especially relevant for XGBoost which has many hyperparamters such as the number of boosting rounds, the learning rate, the maximum tree depth, and the minimum child-node weight. For this reason I will use an additional tuning round for it. It is important to mention that tuning is not based on the final treatment-effect estimate, rather hyperparameters are selected only by predictive performance of the nuisance functions. For each nuisance function j, the selected hyperparameter vector is

$$\hat{\lambda}_j = \arg\min_{\lambda \in \Lambda_j} CV_j(\lambda),$$

where Λ_j denotes the hyperparameter search space and $CV_j(\lambda)$ is the inner cross-validation loss. As Loss for continous outcome regressions the root mean squarred error is used and for binary treatment and instrument models the log-loss is used. The tuning is nested inside the DML procedure. Within each outer cross-fitting fold, the inner cross-validation is performed only on the corresponding training sample. The held-out fold used for evaluating the final orthogonal score is therefore not used for choosing the XGBoost hyperparameters. This avoids tuning directly on the observations that enter the final score evaluation.In the empirical analysis, I apply this tuning step only to the XGBoost Wald-AIPW specification. Especially for the Card framework in which the XGBoost is very reliant on tuning due to the rich covariate specification in the Card and Kitagawa specification. The other learners are used as benchmark nuisance specifications. Tuning will show whether the unstable default XGBoost results are driven by poorly calibrated nuisance predictions.

2 Empirical Application: Military Service and Wages

2.1 Data and Design

> *"Tonight for the first time in 27 years the United States has again started a draft lottery."*
>
> — Roger Mudd, CBS News, 1 December 1969

The economic consequences of military service have long been a central question in the evaluation of veterans' compensation and labour-market policy. In the United States, this question is especially relevant for the Vietnam era, where military service was

politically contested and where the long-run civilian outcomes of veterans remained difficult to assess. A simple comparison between veterans and nonveterans, however, does not provide a credible causal estimate of the effect of military service on earnings. Men who served in the military were not selected at random. They may differ from nonveterans in health, education, ability, labour-market opportunities, or willingness to enlist. As a result, observed earnings differences may reflect both the causal effect of service and the characteristics that selected individuals into military service.
This is the selection problem adressed in Angrist (1990). Key to his idea to overcome it is to use the Vietnam-era draft lottery as a source of quasi random variation in the probability of taking on military service. The lottery did not randomly assigned military service itself, instead, it randomly assigned draft risk.

2.1.1 The Draft Lottery - The instrument

During the Vietnam-era draft lotteries, Random Sequence Numbers (RSNs) were assigned to dates of birth. For each birth cohort, the Department of Defense later announced an induction ceiling. Men with lottery numbers below the relevant cohort-specific ceiling were classified as draft-eligible, while men with higher lottery numbers were not. Formally, draft eligibility can be written as:

$$Z_i \;=\; \mathbf{1}\Big\{RSN_i \leq c_{b(i)}\Big\}, \tag{34}$$

where RSN_i denotes individual i's lottery number, $b(i)$ his birth cohort, and $c_{b(i)}$ the draft-eligibility ceiling for that cohort. Table 2 reports the cohort-specific ceilings for the four birth cohorts in the sample. Hence, draft eligibility did not determine military

Table 2: Draft-eligibility ceilings by birth cohort (Vietnam application)

Birth cohort	Lottery year	Eligibility ceiling
1950	1970	$RSN_i \leq 195$
1951	1971	$RSN_i \leq 125$
1952	1972	$RSN_i \leq 95$
1953	1973	$RSN_i \leq 95$

service perfectly, since some eligible men did not serve and other ineligible men volunteered. It however shift the probability of military service (Relevance). Furthermore, the RSN where randomly assigned to birth dates. As the birth is genuinly an event that is in a random relation to military services especially since partens did not have knowledge at the time that tehre was gonna be a lottery the usage of this instrument can also be seen as as good as randomly assigned. Finally the thought that eligibility would affect later earnings only over military service is an assumption that is plausible to take (Exclusion restriction).

The cohort-specific nature of the lottery matters for the estimators used below. Lottery numbers were assigned randomly within each cohort, but the eligibility ceiling differed across cohorts, so the unconditional probability of draft eligibility varies with birth year:

$$\Pr(Z_i = 1 \mid b(i)) \;=\; \Pr\Big(RSN_i \leq c_{b(i)} \mid b(i)\Big)\,. \tag{35}$$

In the pooled sample, draft eligibility is therefore conditionally random given birth cohort, not unconditionally random. This is why controlling for age is required for the instrument propensity score $p(X_i) = \Pr(Z_i = 1 \mid X_i)$ to be correctly specified.

2.1.2 Sample, Variables, and Design Diagnostics

The used sample in this thesis follows Słoczyński et al. (2025) in revisiting Angrist (1990). Here it especially uses Angrists subsample of the 1984 Survey of Income and Program Participation (SIPP) which contains $N = 3{,}027$ white men born between 1950 and 1953. Table 3 summarises the three core variables of the IV design.

Table 3: IV design variables: Vietnam draft lottery application

Role	Symbol	Definition
Instrument	Z_i	Draft eligibility: $\mathbf{1}\{RSN_i \leq c_{b(i)}\}$; randomly assigned within birth cohort
Treatment	D_i	Vietnam-era veteran status: indicator for military service
Outcome	Y_i	Log hourly wages

Table 4 summarizes the design diagnostics that are of most relevance for the estimator comaprison in the later outcome weights analysis. From the table it can be seen that the assignment probability $\Pr(Z_i = 1) = 0.456$ is close to one half, indicating that draft eligibility was distributed near-evenly in the pooled sample. This reflects the random nature of the RSN assignment within each cohort. The overall veteran rate $\Pr(D_i = 1) = 0.328$ is substantially higher among draft-eligible men ($\Pr(D_i = 1 \mid Z_i = 1) = 0.404$) than among draft-ineligible men ($\Pr(D_i = 1 \mid Z_i = 0) = 0.265$), yielding a first-stage estimate of $\hat{\delta} = 0.139$. Under the IV assumptions, this first stage can be interpreted as the estimated share of compliers, that is, individuals whose veteran status is shifted by draft eligibility.

The first-stage (F)-statistics exceed 26 across all three age specifications. Thus, weak-instrument concerns are not central in this application. At the same time, the nonzero treatment rate $\Pr(D_i = 1 \mid Z_i = 0) = 0.265$ among draft-ineligible men shows that the design exhibits incomplete compliance. Draft eligibility increases the probability of military service, but it does not determine veteran status. This distinction is important

Table 4: Design diagnostics: Vietnam draft lottery (SIPP 1984)

Quantity	Value
Sample size N	3,027
$\Pr(Z_i = 1)$	0.456
$\Pr(D_i = 1)$	0.328
$\Pr(D_i = 1 \mid Z_i = 1)$	0.404
$\Pr(D_i = 1 \mid Z_i = 0)$	0.265
First stage $\hat{\delta}$	0.139
First-stage F, linear age[a]	30.1
First-stage F, cubic age[a]	26.3
First-stage F, saturated age[a]	27.2
Complier share estimate[b]	13.9 %
Compliance pattern	Incomplete

[a] HC1-robust Wald statistic for the coefficient on Z_i in the first-stage regression of D_i on Z_i and the corresponding age controls. With one excluded instrument, this statistic equals the squared robust t-statistic.

for the outcome-weight diagnostics below. The identifying variation comes from the margin shifted by the instrument, not from the full sample. Consequently, the effective sample size of the outcome weights may be much smaller than (N), especially when the weights concentrate on the complier-relevant variation.

Figure 1 illustrates the first stage across age groups. The blue bars show the veteran rate among draft-eligible men ($Z_i = 1$) and the red bars among draft-ineligible men ($Z_i = 0$). Across age groups, veteran rates are higher among draft-eligible men than among draft-ineligible men, providing a visual check of the instrument relevance condition. At the same time, the positive veteran rates among draft-ineligible men show that the design exhibits incomplete compliance.

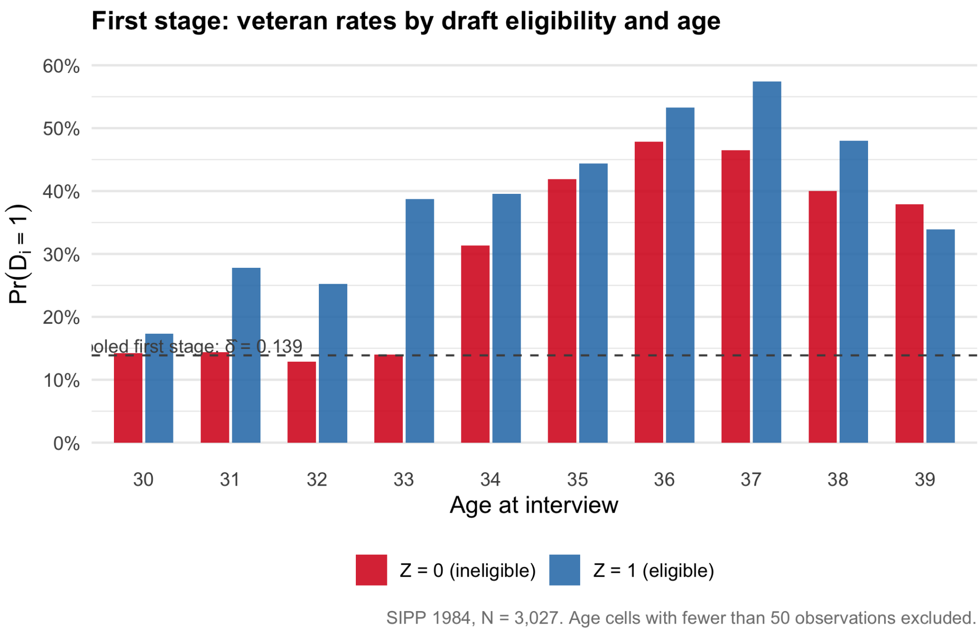

Figure 1: Veteran rates by draft eligibility and age group. Blue bars: $\Pr(D_i = 1 \mid Z_i = 1)$; red bars: $\Pr(D_i = 1 \mid Z_i = 0)$. The gap between bars is the cohort-specific first-stage estimate $\hat{\delta}$. Non-zero red bars confirm incomplete compliance across all age groups. Dashed line: pooled first stage $\hat{\delta} = 0.139$.

Instrument propensity score. A final design feature is the distribution of the estimated instrument propensity score $\hat{p}(X_i) = \Pr(Z_i = 1 \mid X_i)$. The quantity enters all kappa-based estimators and will be therefore relevant for the weight diagnostics below. Extreme or highly dispersed propensity scores can generate more concentrated outcome weights and lower effective sample size. In the Vietnam application, however, the covariate adjustment problem is low-dimensional because (X_i) consists only of age controls. The estimated instrument propensity scores remain centered around

Table 5: Propensity score summary statistics by covariate specification

Specification	Min	P25	Mean	P75	Max	SD
Linear age	0.2810	0.3793	0.4556	0.5258	0.6341	0.1011
Cubic in age	0.1373	0.4110	0.4556	0.5464	0.5501	0.1146
Saturated age	0.2093	0.3707	0.4556	0.5482	0.5671	0.1200

the pooled draft-eligibility rate, with a mean of 0.456 in all three specifications. The main difference across specifications is the amount and shape of variation around this mean. The linear age specification produces fitted values between 0.281 and 0.634, while the cubic and saturated specifications allow more flexibility at the lower end. In particular, the cubic specification reaches a minimum of 0.137, indicating that it assigns substantially lower draft- eligibility probabilities to some age observations. The saturated specification has the largest standard deviation, 0.120, reflecting its discrete age-cell structure. Appendix Figure 6 illustrates this difference visually. Relative to the linear specification, the cubic and saturated specifications produce more clustered fitted values, with the saturated model generating distinct mass points by age group. This makes evere inverse-propensity instability unlikely in this application.

2.2 Point Estimates and Replication

A central finite-sample concern in kappa-based LATE estimation is that some estimators are sensitive to arbitrary transformations of the outcome variable. Słoczyński et al. (2025) provide a a direct illustration of this problem because wages can be measured either in dollars or in cents before taking logarithms

$$\log(100 \cdot wage_i) = \log(wage_i) + \log(100),$$

the two outcomes differ only by an additive constant. Any change in the estimated treatment effect across these two codings therefore reflects a failure of translation invariance in the log-outcome scale. Table 6 uses this recoding exercise to benchmark the kappa estimators, especially with a purpose of not only to reproduce the pattern documented by Słoczyński et al. (2025), but also to establish the point-estimate insights that the outcome-weight diagnostics maybe can explain in the next section.
The normalized estimators are essentially unchanged by the recoding of wages. Across the linear and cubic specifications, ($\hat{\tau}_u^{cb}$),($\hat{\tau}_u^{ml}$), and ($\hat{\tau}_{a,10}$) remain close to the corresponding 2SLS benchmarks. This is the expected behaviour for translation-invariant estimatiors. On the other hand the unnormalized estimators display the opposite pattern. In the linear-age specification, the estimates are negative when wages are measured in cents but close to zero and positive when wages are measured in dollars. $\hat{\tau}_a$ moves from -0.429 to 0.015, an implied gap of 0.444 log points. In the cubic specification, the sign does not change, but the magnitude remains strongly affected by the outcome coding. Interestingly, in the saturated specification all estimators coincide. This is rather due to the fully flexible age-cell specification of the instrument propensity score rather than a general robustness property of the unnormalized estimators. The table therefore motivates the outcome weight analysis that follows.It shows that some estimators react to an additive shift in the log outcome, but the point estimates alone do not identify the mechanical source of the sensitivity. The next section uses this identity to diagnose translation invariance directly through the implied weights.

2.3 Double Machine Learning Comparison

The kappa estimators in the preceding section require an estimate of the instrument propensity score. For $\hat{\tau}_u^{ml}$ and the Abadie-type estimators this is obtained by logistic Maximum Likelihood Estimation (MLE), while $\hat{\tau}_u^{cb}$ uses CBPS moment conditions. The idea here to replace those models with a flexible machine learning nuisance etsimator materially could change the LATE estimate in this design. To assess this, the analysis adds two DML based IV estimators which are the PLR-IV and the WALD-AIPW, both implemented using he OutcomeWeights package of Knaus (2024). The nuisance functions are estimated by generalised random forest with five-fold cross-fitting and automatic hyperparameter tuning.

The linear-age specification is omitted from the DML comparison because it is not a meaningful benchmark for flexible nuisance estimation: with only one continuous covariate, the flexible learner has little scope to improve on the parametric specification. The DML comparison therefore focuses on the cubic and saturated specifications, where the nuisance learner can, in principle, exploit nonlinear age patterns. Relevant benchmarks are provided by the cubic and saturated age specifications, as both supply enough structural variation for the flexible learner to differ, in principle from the other two specifications. Table 6 reports point estimates for the two DML estimators alongside the three normalised kappa estimators and the 2SLS benchmark, for the cubic and saturated specifications and log wages in dollars. Two patterns stand out.

Table 6: Point estimates: DML and kappa estimators (Vietnam)

	Cubic age		Saturated age	
Estimator	Est.	SE	Est.	SE
Panel A: DML				
PLR-IV	0.243	(0.228)	0.242	(0.228)
Wald-AIPW	0.229	(0.231)	0.244	(0.233)
Panel B: Normalised kappa				
$\hat{\tau}_u^{cb}$	0.210	(0.233)	0.241	(0.229)
$\hat{\tau}_u^{ml}$	0.202	(0.235)	0.241	(0.229)
$\hat{\tau}_{a,10}$	0.204	(0.239)	0.241	(0.229)
Panel C: 2SLS benchmark				
2SLS	0.227	(0.229)	0.254	(0.227)

First, the DML estimates are closely aligned with the normalised kappa estimators. In the cubic specification, PLR-IV yields 0.243 and Wald-AIPW yields 0.229, compared with a range of 0.202–0.210 for the normalised kappa estimators and 0.227 for 2SLS. In the saturated specification, PLR-IV gives 0.242 and Wald-AIPW gives 0.244, again consistent with the normalised kappa estimates of 0.241 and the 2SLS benchmark of 0.254. The DML estimates remain statistically and substantively close to the normalized kappa and 2SLS benchmarks, although PLR-IV is somewhat larger than the normalized kappa estimates in the cubic specification. For the PLR-IV it is to take into account though that this framework just estimate the Local Average Treatment Effect under certain conditions. The convergence hence is expected given the design. The Vietnam draft lottery instrument is near-randomly assigned conditional on birth cohort, and the only adjustment variable is age. In this low-dimensional, near-random-assignment setting, the parametric logit and the generalised random forest estimate essentially the same propensity-score surface. The effective sample size and the share of negative weights are virtually identical across all estimators, which confirms that the instrument design rather then than the nuisance learner dirves the weight structure. In this case the primary idea is to suspect that the design dominates the learner.

The generalized random forest based estimates above use a single learner class, a natural question results in assessing whther the convergence result is specific to generalised random forests or whether it holds across fundamentally different nuisance estimators such as parametric, tree ensembles with other packages or boosted sequential learners. These make different approximation choices and carry different theoretical guarantees for the smoother condition $\sum_i \omega_i = 0$. To assess whether learner choice within the DoubleML framework produces different results, Table 6 additionally reports Wald-AIPW estimates from `DoubleMLIIVM` with three alternative nuisance learners: linear regression and logistic classification (the parametric baseline), ranger random forests, and XGBoost. The three learner variants span a range from 0.218 to 0.246. These estimates remain close to the grf Wald-AIPW benchmark of 0.229, the cubic 2SLS estimate of 0.227, and the broader range of normalised kappa estimates reported above. The parametric linear/logistic learner lies at the lower end of the DoubleML range, which is consistent with stronger functional-form restrictions relative to the more flexible tree-based learners. Given the large standard errors and the low-dimensional covariate structure of the Vietnam design, these differences should be interpreted as learner sensitivity checks rather than as distinct empirical conclusions.

2.4 Outcome Weight Diagnostics and Covariate Balance

The point estimate comparison in the prior section showed that unnormalised kappa estimators respond to an additive shift in the log outcome, while normalised estimators and DML do not. Section 2.3 showed that DML and normalised kappa estimators agree numerically. This section explains both findings by looking through the lens of the Outcome weights. Table 7 reports four weight diagnostics for every estimator. The weight sum $\sum_i \omega_i$, the effective sample size $\text{ESS} = 1/\sum_i \omega_i^2$, the share of negative weights, and the maximum absolute weight.

The first column directly operationalises the translation-invariance criterion derived in

Table 7: Outcome weight diagnostics: Vietnam draft lottery

Estimator	$\sum_i \omega_i$	ESS	% neg.	max $\lvert\omega_i\rvert$
Panel A: Cubic age specification				
PLR-IV (DML)	0.000	6	54.4	0.0125
Wald-AIPW (DML)	0.000	5	54.4	0.0176
$\hat{\tau}_u^{cb}$ (kappa)	0.000	5	54.4	0.0254
$\hat{\tau}_u^{ml}$ (kappa)	0.000	5	54.4	0.0281
$\hat{\tau}_{a,10}$ (kappa)	0.000	5	54.4	0.0291
$\hat{\tau}_a$ (kappa)	0.048	5	54.4	0.0289
$\hat{\tau}_t$ (kappa)	0.046	5	54.4	0.0277
$\hat{\tau}_{a,0}$ (kappa)	0.049	5	54.4	0.0291
Panel B: Saturated age specification				
PLR-IV (DML)	0.000	6	54.4	0.0131
Wald-AIPW (DML)	0.000	5	54.4	0.0169
$\hat{\tau}_u^{cb}$ (kappa)	0.000	5	54.4	0.0178
$\hat{\tau}_u^{ml}$ (kappa)	0.000	5	54.4	0.0178
$\hat{\tau}_{a,10}$ (kappa)	0.000	5	54.4	0.0178
$\hat{\tau}_a$ (kappa)	0.000	5	54.4	0.0178
$\hat{\tau}_t$ (kappa)	0.000	5	54.4	0.0178
$\hat{\tau}_{a,0}$ (kappa)	0.000	5	54.4	0.0178

Chapter **??** which can be seen as the necessary and sufficient finite sample condition for translation invariance. In the cubic specification, $\sum_i \omega_i = 0$ holds for PLR-IV, Wald-AIPW, and all three normalised kappa estimators. On the other hand the three unnormalised kappa estimators have weight sums between 0.046 and 0.049, accounting precisely for the instability observed in Table 20. The result in Table 6 extends to the alternative DoubleML learner specifications. The linear/logistic and `ranger` variants both pass the algebraic identity check $\hat{\tau} = \sum_i \omega_i Y_i$ and have $\sum_i \omega_i = 0$, so their recovered outcome weights can be interpreted as translation-invariant Wald-AIPW weights in this application. This is consistent with Knaus (2024), who shows that Wald-AIPW is normalized when the relevant outcome nuisance learners are affine smoothers. XGBoost is different. Although the recovered weights have an almost-zero sum, $\sum_i \omega_i = -3.7 \times 10^{-6}$, they fail the algebraic identity $\hat{\tau} = \sum_i \omega_i Y_i$. This caveat is consistent with Knaus (2024), who notes that boosted trees can violate the affine-smoother condition required for the normalization results.

The saturated specification does not make the unnormalised estimators translation invariant in general. Rather, in this application it estimates the instrument propensity score at the age-cell level, which aligns the sample normalisations of the kappa components closely enough that the implied outcome weights sum to zero. The disappearance of the cents–dollars discrepancy is therefore a feature of this saturated design, not a general robustness property of the unnormalised estimators.

The remaining three columns of Table 7 reveal a striking uniformity. In the cubic specification, every estimator has an ESS of either 5 or 6, a negative-weight share of exactly 54.4%, and a maximum absolute weight below 0.030. These diagnostics are shared in the saturated specification. The uniformity is not a limitation of the diagnostics. It rather reflects a structural feature of the Vietnam design as with a near randomly assigned instrument and a single continuos adjustment variable, the instrument design determines the weight structure almost entirely. The flexible DML model and the kappa estimates that are based on a parametric logistic model and on parametric moment conditions produce virtually the same outcome weights because the propensity- score surface is nearly flat. Hence it can be seen that in the Vietnam application, outcome weight properties are set by design, not the learner.

2.5 Love plots

The weight sum diagnostics establish the translation invariance audit numerically, in this section Love plots follow to assess whether the implied outcome weights also achieve covariate balance in the conventional sense. Figure 2 reports Love plots for all eight estimators in the cubic specification, plotting absolute standardised mean differences (ASMD).The unadjusted ASMD for all three moments lies around 0.5, reflecting the fact that veteran status is not unconditionally randomly assigned. Men who served differ systematically from those who did not in tehir age distribution across cohorts. All eight estimators reduce the adjusted ASMD to approximately 0.05-0.10, well below the conventional 0.1 threshold. This holds uniformly across the DML estimates and the normalised and unnormalised Kappa estimators. The fact that the estimators that fail translation invariance chieve indistinguishable covariate balance from those that pass it once again emphasizes that this is a design specific consequence. This confirms that in this low-dimensional, near-random-assignment setting, the instrument design enforces covariate balance irrespective of normalization choice or learner flexibility.

2.6 Conclusion on Angrist Vietnam Framework

The Vietnam application establishes a solid based behaviour of the unified outcome weights diagnostics in a clean, low dimensional, near-random-assignment design. Several results stand out. Section 2.2 using the cents/dollars recoding reproduces the central finding of Słoczyński et al. (2025).Normalised kappa estimators are invariant to additive outcome shifts, while unnormalised estimators are not, with the linear-age specification reversing sign entirely. In the case of the staurated specification all estimators collapse to a common value (0.241). As Słoczyński et al. (2025) note his as a demonstration that fully flexible propensity-score specifications can remove finite-sample pathologies. Moreover, using the *Outcome Weights* package provided by Knaus

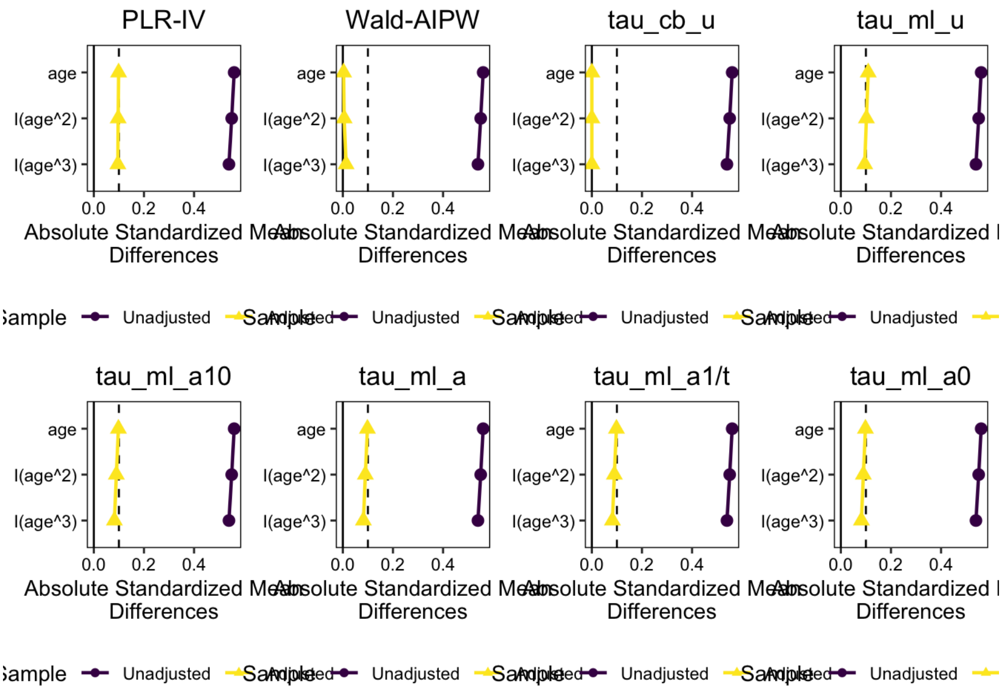

Figure 2: Dark circles: unadjusted sample. Yellow triangles: weight-adjusted. Dashed line: 0.1 threshold. balance assessed using `love.plot()` from COBALT (?).

(2024) it is can be seen that the identity $\hat{\tau} = \sum_i \omega_i Y_i$ holds and point estimates, effective sample sizes, and negative-weight shares are virtually identical across all kappa and DML estimators and across grf, ranger, and XGBoost learners. It is to mention that when using Wald-AIPW with gradient-boosted trees the affine-smoother condition fails. This caveat is also documented by Knaus (2024) as XGBoost does not produce an exact smoother matrix not guaranteeing that the weights sum up to 0 even with the four required hyperparameters set.

Apart from the application of the theoretical concepts a central finding is that in this design the instrument structure, not the choice of estimator or learner, determines the weight properties, this can be shown as the weight diagnostics are virtually identical across all estimators. . The four Wald-AIPW variants (grf 0.229, linear+logit 0.218, ranger 0.246, XGBoost 0.246) span only 0.03 log points, all consistent with the normalised kappa range of 0.20–0.21. The covariate balance is achieved equally by all estimators as Unadjusted ASMD 0.5 on all three age moments all eight estimators reduce it below 0.1. The estimators that FAIL translation invariance achieve the same balance as those that pass it, emphasizing that the balance is rather set by design and nor by normalization choice. Finally in the saturated design the parametric learner is the one that breaks as the linear and logistic Wald-AIPW cannot produce outcome weights in the saturated specification, while flexible learners can handle it without issue. Hence, the Vietnam design can be seen as a benchmark precisely because the diagnostics are almost too clean. In the next section the Card (1993) application introduces a richer covariate-adjustment problem and an observational instrument, to detect whether the diagnistcs begin to differentiate across estimators.

3 Empirical Applications: Card (1995)

One of the central empirical questions in labour economics is the effect of education on earnings. Human capital tradition often displays schooling as an investment that raises productivity and is therefore expected to increase later wages. The difficulty is that schooling is not random assigned. Individuals who acquire more education may differ systematically in ability, motivation, family background, or labour-market expectations, so a simple regression of wages on schooling need not recover a causal return. This identification problem motivated a large literature on the econometrics of schooling returns. Griliches (1977) emphasized that estimates of the return to schooling are affected by omitted ability and measurement error, making both the direction and the magnitude of bias theoretically ambiguous. Later work therefore turned to quasi-experimental sources of variation in education. Angrist and Krueger (1991), for example, use compulsory-schooling laws interacted with quarter of birth as an instrument for completed education. The college-proximity design of Card (1993) belongs to this causal-identification tradition. It uses geographic proximity to a four year college as an instrument for schooling.

The Card application is useful for this thesis because it changes the empirical design relative to the Vietnam draft lottery. In the Vietnam application, the instrument was close to randomly assigned conditional on age, and the covariate adjustment problem was deliberately low-dimensional. In the Card application, by contrast, college proximity is observational. The credibility of the instrument depends on conditioning on individual, family, regional, and labour-market characteristics. The instrument propensity score is therefore no longer nearly flat, and the choice of covariate specification becomes part of the empirical problem. This makes it a natural second application, as it tests whether the outcome-weight diagnostics remain uniform once the instrument is observational and the covariate structure is richer.

3.1 Data

The analysis follows Słoczyński et al. (2025) in revisiting the Card college-proximity application using the National Longitudinal Survey of Young Men. The sample contains 3,010 men with valid wage and education information who were interviewed in 1976. The outcome is log hourly wages.

3.1.1 College Proximity: The instrument

The instrument in this design is an indicator for whether an individual grew up near a four year college in 1966. The economic mechanism behind is a cost of attendance channel. Individuals that live close to a four-year college could attend without bearing the full cost of relocation or long-distance commuting. Proximity should therefore increase the probability of post-secondary schooling, especially for individuals whose schooling choices are sensitive to the direct and indirect costs of college attendance. Unlike draft eligibility in the Vietnam application, college proximity is not randomized by design. In the draft-lottery setting, conditional on birth cohort, the instrument is generated by the random assignment of lottery numbers to dates of birth. The source of variation is therefore institutional and close to experimental. In the Card application, by contrast, the instrument is determined by where individuals grew up. Living near a four-year college is related to residential location and therefore potentially to family background, region, urban status, school quality, and local labour-market conditions. These factors may also affect later wages directly. Formally, the instrument must shift schooling choices, satisfy exclusion from the wage equation, and be conditionally independent of potential earnings given the covariates. Under these assumptions, the IV estimand is a local average treatment effect for individuals whose schooling decision is shifted by access to a nearby college.

Table 8: IV design variables: Card application

Role	Symbol	Definition
Instrument	Z_i	Near a four-year college in 1966,
Treatment 1	D_{1i}	Some college attendance, $1\{\texttt{educ}_i > 12\}$
Treatment 2	D_{2i}	College completion, $1\{\texttt{educ}_i \geq 16\}$
Outcome	Y_i	Log hourly wage

3.1.2 Treatment Definitions and Outcomes

Card's original application studies the effect of completed years of schooling on earnings (Card, 1993). The kappa estimators considered in this thesis, however, are defined for a binary treatment. I therefore use two binary schooling margins. The first treatment is some college attendance, which indicates whether an individual obtained any schooling beyond high school. The second treatment is college completion, which indicates whether individual completed at least sixteen years of education. The two treatment margins correspond to different complier groups. The first group of compliers are individuals whose decision to continue beyond high school is shifted by growing up near a four-year college. The second group of compliers are individuals whose probability of completing college is shifted by the same instrument. The resulting Local Average Treatment Effects should therefore not be interpreted as the average return to one additional year of schooling in Card's original continuous-treatment specification (Card, 1993). They are binary-treatment Local Average Treatment Effects for the schooling margin affected by college proximity.

3.1.3 Covariate Specifications

For the reason that college proximity is not randomly assigned, the covariates play a more central role in the Card application than in the Vietnam draft-lottery design. In Vietnam, conditioning on age mainly reflected the cohort-specific lottery rules. **In the Card application, by contrast, the covariates define the conditional exogeneity assumption itself.** The choice of covariate specification is therefore not merely a robustness detail, but a part of the identifying design. Paralleled Słoczyński et al. (2025) I use two covariate specifications. The first follows Card (1993) and is the richer baseline specification. It includes experience, experience squared, indicators for Black, living in the South, living in an SMSA in 1976, living in an SMSA in 1966, and regional indicators.This specification controls for individual labour-market experience and geographic differences in college access and wage levels. It is therefore very close to Card's original design. The second specification follows Kitagawa (2015) and

Table 9: Covariate specifications in the Card application

Covariate	Card specification	Kitagawa specification
Experience, `exper`	Yes	No
Experience squared, `expersq`	Yes	No
Black, `black`	Yes	Yes
South in 1976, `south`	Yes	Yes
SMSA in 1976, `smsa`	Yes	Yes
SMSA in 1966, `smsa66`	Yes	Yes
South in 1966, `south66`	No	Yes
Region indicators, `reg661–reg668`	Yes	No

is more parsimonious. It includes indicators for Black, living in the South in 1976, living in an SMSA in 1976, living in an SMSA in 1966, and living in the South in 1966. This lower-dimensional specification is useful because it has been used in the binary-treatment IV literature and because it provides a direct check of how sensitive the outcome-weight diagnostics are to the richness of the covariate adjustment set. The comparison between the two will be central for the present analysis. In the next sections a few descriptive analysis will be depicted with the purpose of not merley describe the sampel but to evaluate whether the instrument has sufficient first-stage relevance and whether overlap in the instrument propensity score is plausible.

3.2 Descriptives

Table 10 reports important descriptives such as the unconditional treatment shares and the treatment rates by instrument status. This is usefully as it shows the amount of variation available and especially how strongly the instrument shifts the treatment probability under monotonicity. It can be seen how there are clear discrepancies between our first stages what depicts clearly the strength of the first stage. For the some-college margin, the probability of treatment is 54.4 percent among individuals who grew up near a four-year college and 42.2 percent among those who did not. The raw first-stage gap is therefore 12.2 percentage points. On the other hand, for the college-completion margin, the first stage is substantially smaller. The probability of completing at least sixteen years of schooling is 29.3 percent among individuals near a four-year college and 22.5 percent among those farther away. The implied raw first stage is only 6.9 percentage points. This indicates that the stricter treatment margin is identified from a much smaller complier group. This shows that the empirical design contains less identifying variation for college completion than for some college attendance. Furtheremore, the table shows that neither treatment margin corresponds to one-sided compliance, as there are substantial always-taker and never-taker shares. Next to the raw first stage the the covariate-adjusted first stage is very important,

Table 10: Raw compliance diagnostics in the Card application

	`somecol`	`educ16`
$\Pr(D = 1)$	0.505	0.271
$\Pr(D = 1 \mid Z = 1)$	0.544	0.293
$\Pr(D = 1 \mid Z = 0)$	0.422	0.225
Raw first stage / complier share	0.122	0.069
Always-taker share	0.422	0.225
Never-taker share	0.456	0.707

hence Table 11 compares the covariate adjusted first stage for the two treatment specifications using both covariate sets. It can be clearly seen for `somecol`, how once we include Card or Kitagawa covariates are included, the first stage falls to about 6.4–6.5 percentage points. Once tge part of the raw relationship between college proximity and some college attendance is explained by the covariates, the value of the first stage nearly drops by the half. The pattern is more concerning for `educ16`. The raw first stage is already smaller, at 6.9 percentage points, and the covariate-adjusted first stage falls further to 3.0 percentage points under the Card specification and 3.8 percentage points under the Kitagawa specification.

The corresponding first-stage F-statistics are only 3.5 and 3.9. This indicates that college proximity provides only limited identifying variation for the college-completion margin once the covariates are held fixed.

This gives already a central interpretation for the later estimator comparisons implying that the `somecol` estimates are based on a not very strong first stage. By contrast, the `educ16` estimates rely on a much weaker first stage. Large Wald contrasts or unstable weighting estimates for `educ16` should therefore not be over-interpreted.

Table 11: First stage, reduced form, and Wald contrasts in the Card application

Treatment	Specification	First stage $\hat{\delta}$	F-statistic	Reduced form $\hat{\gamma}$	Wald $\hat{\gamma}/\hat{\delta}$
somecol	Card	0.0636	12.5	0.0421	0.6613
somecol	Kitagawa	0.0653	9.0	0.0375	0.5748
educ16	Card	0.0302	3.5	0.0421	1.3915
educ16	Kitagawa	0.0379	3.9	0.0375	0.9909

A final descriptive diagnostic that is especially interesting for the current framework is the distribution of the fitted instrument propensity score. This will especially matter for the Kappa estimators as they use inverse propensity scores. Extreme fitted probabilities can therefore create large weights and reduce the effective sample size in the later outcome-weight diagnostics. Table 12 summarizes the fitted propensity scores for the Card and Kitagawa specifications. The Card specification produces slightly more dispersion, with fitted values ranging from 0.172 to 0.952 and a standard deviation of 0.236. The Kitagawa specification ranges from 0.253 to 0.907 and has a similar but slightly smaller standard deviation of 0.228. Figure 3 provides a more detailed view

Table 12: Instrument propensity-score summary in the Card application

Specification	Min	P05	P25	Mean	P75	P95	Max	SD	Near 0/1
Card	0.1724	0.2459	0.4376	0.6821	0.8726	0.9341	0.9517	0.2356	0.0056
Kitagawa	0.2532	0.2532	0.4554	0.6821	0.8783	0.8783	0.9067	0.2280	0.0000

by plotting the propensity-score distribution separately for the two instrument groups. Here the most relevant feature is the upper tail. Specifically in the Card specification, many individuals have high predicted probabilities of living near a four-year college, but relatively few observations with $(Z_i = 0)$ appear in that same region. These observations will later turn out to be important as when $(\hat{p}(X_i))$ is close to one, the inverse term $(1/(1 - \hat{p}(X_i)))$becomes large for individuals with $(Z_i = 0)$. They can be seen as individuals whose covariates make them look likely to live near a college but who in fact did not, which are rare comparison observations. The Kitagawa specification is more discrete because it uses fewer covariates, but it shows a similar qualitative pattern. Most mass is concentrated above the pooled instrument share, and the relevant scarcity concerns occur mainly for $(Z_i = 0)$ observations in high-propensity regions.

3.3 Point Estimates and Replication

As shown in the previous descriptives section and grounded by the fact that college proximity is not randomly assigned, the Card application is a more demanding IV design than the Vietnam draft-lottery application. Hence, next it is to examine whether

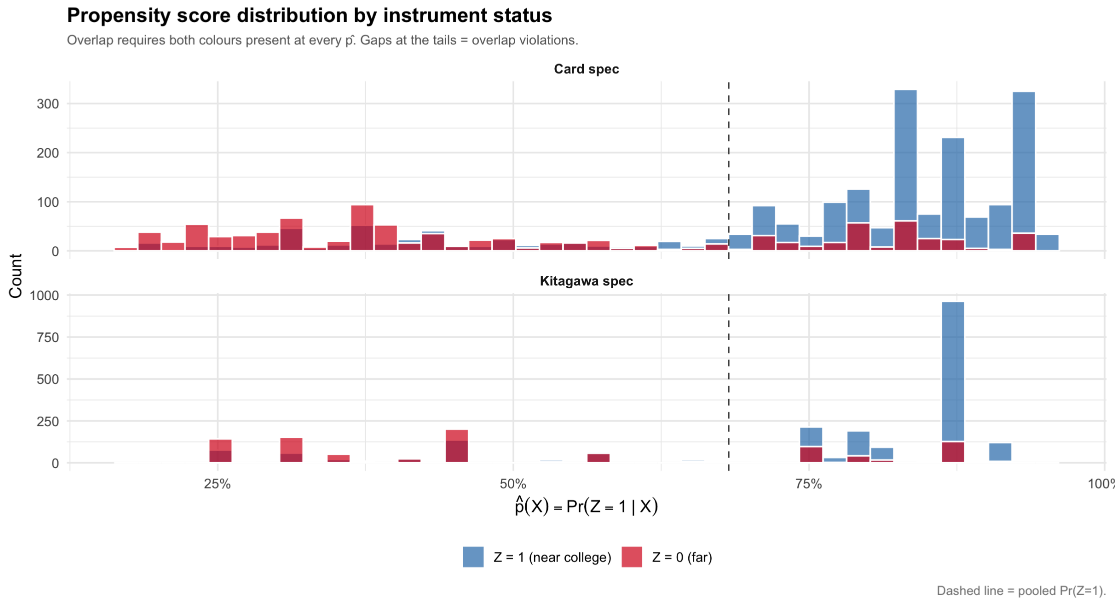

Figure 3: Distribution of the fitted instrument propensity score by instrument status

the design features translate into the point estimates documented by Słoczyński et al. (2025). The purpose of the replication is that it provides a benchmark against which the later outcome-weight diagnostics can be interpreted. If the point estimates already differ strongly across estimators or covariate specifications, later weight diagnostics may can help to explain this instability. Table 13 reports the main point estimates for log wages measured in dollars. The point estimates for log wages measured in cents will then be explored in the Translation invariance section, in which the reproduction section will be expanded using log wages in cents and a translation invariance pipeline will be designed.

First, taking a look at the `somecol` specification it is visible that the normalized kappa estimates are moderate and relatively close across the two covariate specifications. Under the Card specification, the normalized estimates range from 0.331 to 0.376. Under the Kitagawa specification, they range from 0.293 to 0.356. Thus, once the estimator is normalized, changing the covariate set affects the estimates but does not lead to large qualitative changes. On the other hand, the unnormalized estimators behave differently.They seem close to the normalized estimates under the Card specification, but become much larger under the Kitagawa specification. For example, $\hat{\tau}_a^{ml}$ increases from 0.170 under the Card specification to 0.842 under the Kitagawa specification, and $\hat{\tau}_{a,0}^{ml}$ increases from 0.154 to 1.066. Important to mention that all the estimators are still quite far from the benchamrk 2SLS estimator.

The difference is even clearer for the `educ16` margin. he normalized kappa estimates remain between 0.586 and 0.853, although with large standard errors. By contrast, the unnormalized estimates become substantially larger under the Kitagawa specification, reaching 1.617 for $\hat{\tau}_a^{ml}$ and 2.712 for $\hat{\tau}_{a,0}^{ml}$. This is consistent with the descriptive evidence that showd that the adjusted first stage for college completion is weak under both specifications, especially for the college completition application. Estimators hence already reflect the combination of a weak first stage and estimator-specific normalization behavior.

Rather than comparing point estimates alone, the following sections use the outcome-weight representation to examine which observations receive influential weights, whether the final weights are normalized, and whether the resulting weighted comparison improves covariate balance.

Table 13: Kappa and 2SLS point estimates in the Card application

	`somecol`		`educ16`	
Estimator	Card	Kitagawa	Card	Kitagawa
Panel A: 2SLS				
2SLS	0.661**	0.575*	1.392*	0.991
	(0.295)	(0.308)	(0.801)	(0.611)
Panel B: Normalized kappa estimators				
$\hat{\tau}_u^{cb}$	0.376*	0.331	0.853	0.588
	(0.223)	(0.236)	(0.549)	(0.433)
$\hat{\tau}_u^{ml}$	0.331	0.356	0.619	0.628
	(0.202)	(0.244)	(0.387)	(0.448)
$\hat{\tau}_{a,10}^{ml}$	0.346*	0.293	0.586*	0.836
	(0.200)	(0.252)	(0.356)	(0.821)
Panel C: Unnormalized kappa estimators				
$\hat{\tau}_a^{ml}$	0.170	0.842**	0.315	1.617*
	(0.370)	(0.362)	(0.696)	(0.891)
$\hat{\tau}_t^{ml}$	0.171	0.769**	0.319	1.367**
	(0.367)	(0.308)	(0.687)	(0.648)
$\hat{\tau}_{a,0}^{ml}$	0.154	1.066*	0.266	2.712
	(0.354)	(0.574)	(0.639)	(2.577)

Notes: The table reports point estimates for log wages measured in dollars. Standard errors are shown in parentheses. Stars denote significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

3.4 Double Machine Learning Estimates

A in Section 2.3 I complement the classical IV and kappa estimates with double machine learning estimators. Again I consider the PLR-IV, which is based on a partially linear IV score. Moroever, the Wald-AIPW, which combines a binary-instrument Wald contrast with augmented inverse-propensity components Table 14 shows a clear separation between PLR-IV and Wald-AIPW. PLR-IV is close to the conventional 2SLS benchmark. For the `somecol` margin, PLR-IV gives 0.681 under the Card specification and 0.593 under the Kitagawa specification, almost identical to the corresponding 2SLS estimates. The same pattern appears for `educ16`, where PLR-IV remains close to the larger 2SLS estimates. This is consistent with the interpretation of PLR-IV as a flexible version of a partially linear IV, rather than as a direct analogue of the normalized kappa LATE estimators. On the other hand, the Wald AIPW behaves different. The grf and ranger implementations are substantially smaller than PLR-IV and closer to the normalized kappa benchmark. For `somecol`, both forest-based Wald-AIPW estimates lie around 0.26–0.29, compared with normalized kappa estimates of 0.331 and 0.356. For `educ16`, the forest-based Wald-AIPW estimates are around 0.49–0.54, again below PLR-IV and 2SLS but close to the normalized kappa estimates. These estimates are very close to the $\hat{\tau}_u^{ml}$ but more conservative then the PLR-IV. The parametric linear/logit Wald-AIPW estimator provides an intermediate benchmark. For `somecol`, it is close to the normalized kappa and forest-based Wald-AIPW estimates. For `educ16`, however, it is larger under the Card specification. This could be due to the fact that the parametric learner imposes stronger functional-form restrictions on the nuisance functions. The XGBoost row is qualitatively different. Its estimates are not only larger or smaller than the other DML estimates, but they are outside the empirical range of all other estimators and even change sign. This again showcases evidence that the Wald-AIPW score is highly sensitive to the nuisance learner in this application. The score depends on several estimated nuisance components, including instrument propensities and conditional outcome and treatment regressions. In a weak-first-stage design, small errors in these components can strongly affect the final Wald ratio. Sequential boosting can amplify this problem if the fitted nuisance functions are poorly calibrated in regions with limited overlap.

Table 14: DML-IV estimates in the Card application

	`somecol`		`educ16`	
Estimator	Card	Kitagawa	Card	Kitagawa
2SLS	0.661	0.575	1.392	0.991
$\hat{\tau}_u^{ml}$ (kappa)	0.331	0.356	0.619	0.628
PLR-IV (grf / OutcomeWeights)	0.681	0.593	1.140	1.030
Wald-AIPW (grf / OutcomeWeights)	0.279	0.292	0.530	0.535
Wald-AIPW (linear/logit)	0.399	0.311	0.884	0.534
Wald-AIPW (ranger)	0.262	0.294	0.493	0.514
Wald-AIPW (XGBoost)	-13.416	-0.113	-3.198	-3.656

3.4.1 Nested XGBoost tuning as a Robustness check

The instability of the default XGBoost specification motivates a nested tuning robustness check. The purpose of this exercise is not to search for a treatment-effect estimate that looks plausible. Instead, the XGBoost hyperparameters are selected only by predictive nuisance performance inside the DML procedure. The final treatment-effect estimate is not used as a tuning criterion, as for the continuous outcome nuisance, tuning minimizes root mean squared error and for the binary treatment and instrument nuisances, tuning minimizes log-loss. Table 15 reports the resulting untuned and tuned XGBoost Wald-AIPW estimates. The untuned results reproduce the extreme XGBoost rows in Table 15. After tuning, the estimates move back into the range of the other Wald-AIPW and normalized kappa estimates. For the Card specification, the somecol estimate changes from (-13.416) to (0.352), while the educ16 estimate changes from (-3.198) to (0.627). The same pattern appears under the Kitagawa specification, where the tuned estimates are (0.258) for somecol and (0.475) for educ16. The tuning results suggest that the extreme default XGBoost estimates are largely driven by nuisance calibration rather than by a stable causal signal. In the Card specification, the default XGBoost learner produces many boundary propensity predictions, while the tuned specification removes them under the chosen threshold. This supports the interpretation that the original XGBoost row is a diagnostic warning about learner instability. At the same time, the tuned XGBoost estimates should still be interpreted as a sensitivity check rather than as the preferred estimator. Overall, flexible nuisance estimation does not remove the empirical difficulty of this design. PLR-IV remains close to 2SLS, while Wald-AIPW with grf, ranger, and the parametric learner is closer to the normalized kappa estimates. The default XGBoost specification is the exception and is best interpreted as a learner-instability diagnostic. The nested tuning exercise shows that this instability is strongly reduced when XGBoost is selected by nuisance-prediction loss.

Table 15: Nested XGBoost nuisance tuning: Wald-AIPW estimates

Cell	XGB untuned	XGB tuned	Propensity boundary share
Card–somecol	(-13.416) ((14.411))	(0.352) ((0.228))	(0.2595 → 0.0000)
Card–educ16	(-3.198) ((4.054))	(0.627) ((0.420))	(0.2595 → 0.0000)
Kitagawa–somecol	(-0.113) ((0.627))	(0.258) ((0.269))	(0.0017 → 0.0000)
Kitagawa–educ16	(-3.656) ((100.914))	(0.475) ((0.492))	(0.0017 → 0.0000)

3.5 Outcome Weight Diagnostics: Dispersion, Extrapolation, and Normalization

The previous sections compared point estimates. I now turn to the outcome weights behind these estimates.

3.5.1 Overall Dispersion and Extrapolation

Table 16 reports the main dispersion diagnostics for a selected set of estimators. Paralleled to the Vietnam framework I include PLR-IV and Wald-AIPW as the two DML-IV estimators, ($\hat{\tau}^{cb}u$) and ($\hat{\tau}^{ml}u$) as normalized kappa benchmarks, and ($\hat{\tau}_t^{ml}$) as a representative unnormalized kappa estimator. Because the outcome weights are signed, I use the modified effective sample size as the main dispersion measure. The usual Kish effective sample size is designed for nonnegative normalized weights and can become difficult to interpret for signed contrasts. This measure remains interpretable for signed weights and reduces to the usual Kish expression when all weights are nonnegative. The first result is that all non-XGBoost estimators shown in Table 16 retain a substantial modified effective sample size. Across the four design cells, (ESS_{mod}) remains between approximately 1345 and 1954 for the DML estimators and between approximately 1561 and 1786 for the kappa estimators. Relative to the nominal sample size of 3010, the Card application therefore uses a smaller but still substantial effective sample. Furtheremore when comparing the 2 treatment specifications it can be seen that the `educ16` margin generates more concentrated weights than the `somecol` margin. This is visible in the maximum absolute weights and in the top-ten-percent mass. Under the Card specification, the Wald-AIPW maximum weight increases from 0.146 for `somecol` to 0.277 for `educ16`. Similarly, ($\hat{\tau}_u^{cb}$) increases from 0.087 to 0.197. The same pattern appears under the Kitagawa specification, although less sharply. This again underlines the descriptive weak first stage diagnostic as college completion is weakly shifted by the instrument, whihch causes the estimators to require more concentrated weights to construct the complier contrast. Furtheremore, the share of negative weights is remarkably stable across the main estimators. It is close to 32 percent in almost every non-XGBoost case. Thus, the negative-weight share itself is not the main source of estimator differentiation in the Card application. What varies more meaningfully is the magnitude of the largest weights and the amount of absolute reweighting. Negative weights are therefore best interpreted together with weight magnitude and effective sample size,instead of a standalone failure criterion.

3.5.2 Treated-Control Normalization

After the examining the dispersion and extrapolation in the weights, a different question is whether the outcome weights form a normalized treated-minus-control comparison. Under full normalization both quantities equal one.

Table 17 shows that the normalized kappa estimator ($\hat{\tau}_u^{ml}$) is exactly normalized in all four design cells. Its treated mass equals one and its sign-flipped untreated mass also equals one. Due to its structure of a normalized weighted treated mean minus a normalized weighted untreated mean, ($\hat{\tau}_u^{ml}$) can be seen as the cleanest finite-sample benchmark. The DML estimators are close to this benchmark but not identical to it. PLR-IV is very close to one in all four cells. Wald-AIPW is also close, although slightly below one in the Card specification and slightly above one in the Kitagawa `educ16`

Table 16: Outcome-weight dispersion diagnostics in the Card application

Cell	Estimator	Min	Max	% Neg.	Σ_{Top10}	$\sum\lvert\omega\rvert$	ESS_{mod}
Panel A: Card specification, `somecol`							
Card–`somecol`	PLR-IV (DML)	-0.033	0.032	33.1	6.03	30.11	1819
Card–`somecol`	Wald-AIPW (DML)	-0.146	0.039	31.8	3.03	20.70	1345
Card–`somecol`	$\hat{\tau}_u^{cb}$	-0.087	0.022	31.8	3.48	23.46	1609
Card–`somecol`	$\hat{\tau}_u^{ml}$	-0.052	0.020	31.8	2.94	20.28	1569
Card–`somecol`	$\hat{\tau}_t^{ml}$	-0.055	0.020	31.8	3.07	21.31	1561
Panel B: Card specification, `educ16`							
Card–`educ16`	PLR-IV (DML)	-0.055	0.053	33.1	10.10	50.42	1819
Card–`educ16`	Wald-AIPW (DML)	-0.277	0.074	31.8	5.76	39.31	1345
Card–`educ16`	$\hat{\tau}_u^{cb}$	-0.197	0.051	31.8	7.90	53.20	1609
Card–`educ16`	$\hat{\tau}_u^{ml}$	-0.097	0.037	31.8	5.50	37.96	1569
Card–`educ16`	$\hat{\tau}_t^{ml}$	-0.103	0.038	31.8	5.75	39.86	1561
Panel C: Kitagawa specification, `somecol`							
Kitagawa–`somecol`	PLR-IV (DML)	-0.033	0.025	31.8	6.51	33.33	1954
Kitagawa–`somecol`	Wald-AIPW (DML)	-0.065	0.015	31.8	3.51	24.57	1631
Kitagawa–`somecol`	$\hat{\tau}_u^{cb}$	-0.046	0.016	31.8	3.44	23.91	1743
Kitagawa–`somecol`	$\hat{\tau}_u^{ml}$	-0.044	0.016	31.8	3.53	24.49	1764
Kitagawa–`somecol`	$\hat{\tau}_t^{ml}$	-0.038	0.014	31.8	3.10	21.21	1786
Panel D: Kitagawa specification, `educ16`							
Kitagawa–`educ16`	PLR-IV (DML)	-0.057	0.043	31.8	11.32	57.95	1954
Kitagawa–`educ16`	Wald-AIPW (DML)	-0.119	0.028	31.8	6.42	44.98	1631
Kitagawa–`educ16`	$\hat{\tau}_u^{cb}$	-0.081	0.028	31.8	6.12	42.50	1743
Kitagawa–`educ16`	$\hat{\tau}_u^{ml}$	-0.078	0.028	31.8	6.23	43.22	1764
Kitagawa–`educ16`	$\hat{\tau}_t^{ml}$	-0.067	0.025	31.8	5.51	37.72	1786

Table 17: Treated-control normalization of outcome weights

Cell	PLR-IV	Wald-AIPW	Linear/logit	Ranger	XGBoost	$\hat{\tau}_u^{ml}$	$\hat{\tau}_t^{ml}$
Card–`somecol`	1.00/1.00	0.97/0.97	1.00/1.00	0.98/0.98	-7.31/-6.03	1.00/1.00	1.00/1.11
Card–`educ16`	0.96/0.96	0.94/0.94	0.92/0.92	0.95/0.95	-2.19/-1.88	1.00/1.00	1.00/1.20
Kitagawa–`somecol`	1.00/1.00	0.99/0.99	1.00/1.00	1.00/1.00	0.43/0.49	1.00/1.00	1.00/0.72
Kitagawa–`educ16`	1.01/1.01	1.01/1.01	0.99/0.99	1.00/1.00	0.75/2.75	1.00/1.00	1.00/0.50

case. The parametric linear/logit and ranger versions behave similarly. This is consistent with the idea that these implementations produce approximately normalized contrasts, but not the exact finite-sample normalization imposed by the normalized kappa formulas. The unnormalized kappa estimator ($\hat{\tau}_t^{ml}$) illustrates why treated-control normalization is informative. Its treated mass is one by construction, but the untreated mass is not. Under the Kitagawa specification, the sign-flipped untreated mass falls to 0.72 for `somecol` and 0.50 for `educ16`. This is the untreated-unnormalized case mentioned in Knaus (2024), where the treated component is normalized by construction, while the untreated component is not separately rescaled to minus one. The XGBoost row is qualitatively different from all other rows. In the Card specification, the treated and untreated masses are negative. In the Kitagawa `educ16` case, the untreated mass rises to 2.75. This confirms that the XGBoost estimates are not just different point estimates. They are generated by a fundamentally different and unstable weight structure.

3.5.3 Learner Sensitivity within Wald-AIPW

When taking a look at the learner comparison in the max absolute weights and in the modified effective sample size it is visible for the ranger specification that its modified ESS and maximum absolute weights are close to the grf and linear/logit implementations. For example, in the Kitagawa specification, ranger has (ESS_{mod}) above 1600 in both treatment margins, and its maximum absolute weights remain below 0.10. The XGBoost is the clear exception. In the Card specification, (ESS_{mod})falls to 74 for both treatment margins. The maximum absolute weight reaches 20.03 for `somecol` and 4.77 for `educ16`. In the Kitagawa `educ16` case, the maximum absolute weight reaches 22.04. These values are orders of magnitude larger than the corresponding linear/logit and ranger weights. This underlines that the extreme XGBoost point estimates are therefore not an isolated numerical accident, but rather the direct consequence of an unstable weight geometry.
This can be seen as a substantive implemenation finding as it indicates that the Wald-AIPW score can become highly sensitive to the nuisance learner in a weak-first-stage, observational IV design. The Card application combines a nonrandom instrument, relevant covariate imbalance, and limited identifying variation for college completion. In this environment, highly adaptive nuisance estimation can generate extreme residualized or inverse-propensity components, which then propagate into the final outcome weights. Overall, the outcome-weight diagnostics clarify the point-estimate results. PLR-IV and the valid Wald-AIPW implementations use sizeable effective samples and have treated-control masses close to one. Normalized kappa estimators provide the exact finite-sample normalization benchmark. The unnormalized kappa estimator differs mainly in the treated-control mass structure, not in the negative-weight share.

XGBoost is the only case where the weight structure itself becomes pathological.

Table 18: Learner sensitivity of Wald-AIPW outcome weights

Cell	ESS_{mod}			$\max_i \lvert\omega_i\rvert$		
	Linear/logit	Ranger	XGBoost	Linear/logit	Ranger	XGBoost
Card–`somecol`	1555	1309	74	0.081	0.135	20.028
Card–`educ16`	1555	1309	74	0.179	0.254	4.774
Kitagawa–`somecol`	1724	1681	435	0.048	0.057	0.683
Kitagawa–`educ16`	1724	1681	435	0.082	0.099	22.039

3.6 Translation invariance

As in the Vietnam application, wages will be later measured either in cents or in dollars before taking logarithms, so the two outcome variables differ by the additive constant (log(100)). This preserves the same translation-invariance test used in the preceding chapter.

3.7 Love Plots

After having shown whether the weights are concentrated and normalized, it is now to determine for this particular specification whether treated and untreated components of the final outcome-weight contrast actually look balanced in covariates. The balance diagnostics are shown using Love plots. Each plot reports absolute standardized mean differences before and after applying the estimator-specific outcome weights. The unadjusted points describe the raw imbalance between treated and untreated observations. The adjusted points describe the imbalance induced by the final outcome weights. The dashed vertical line marks the conventional threshold of 0.1. The love plots will focus on the Card specification and the `educ16`. This is the most informative case because the descriptive diagnostics already showed that college completion has the weakest first stage. The weight diagnostics also showed larger maximum weights and stronger concentration for `educ16` than for `somecol`. The Love plots therefore test whether this more fragile weight structure translates into weaker covariate balance. Figure 4 compares PLR-IV, Wald-AIPW, and the kappa estimators. The unadjusted imbalance is substantial, especially for experience, experience squared, race, region, and urban-status variables. This confirms again that the Card application is not a near-random-assignment design like the Vietnam draft lottery.After weighting, the estimators behave very differently. The CBPS-based normalized kappa estimator ($\hat{\tau}_u^{cb}$) achieves the strongest balance improvement, with adjusted standardized mean differences close to zero across almost all displayed covariates. This is consistent with the role of CBPS as a balancing propensity-score method. The other estimators do not uniformly achieve the same balance. PLR-IV and Wald-AIPW reduce some of the largest raw imbalances, but several adjusted regional differences remain above the 0.1 threshold. The ML-based normalized kappa estimator ($\hat{\tau}_u^{ml}$) also satisfies the normalization properties discussed above, but it does not automatically deliver covariate balance. In particular, several regional indicators remain strongly imbalanced after weighting. Figure 5 examines learner sensitivity within Wald-AIPW. The linear/logit implementation provides the cleanest balance among the DoubleML learner variants. Ranger performs reasonably but leaves larger adjusted imbalances for some regional variables. XGBoost is the clear exception. Its adjusted standardized mean differences are large for several covariates, including urban-status and regional indicators. This mirrors the

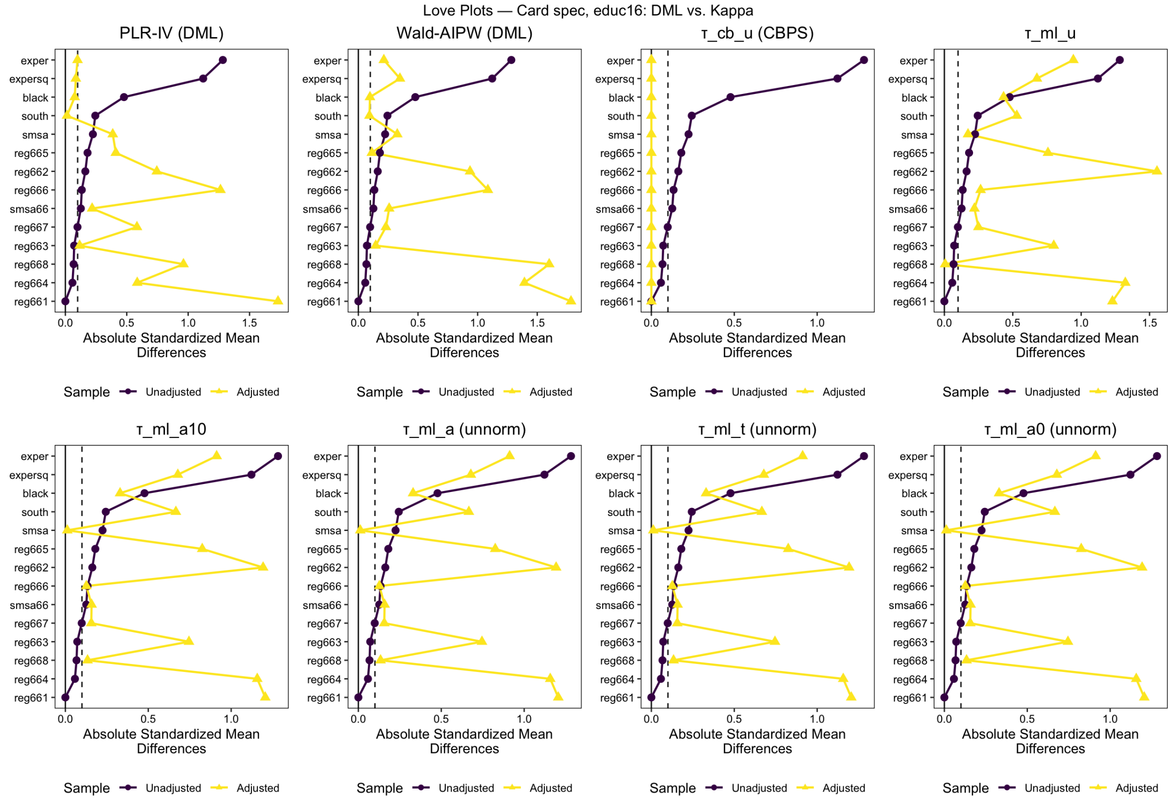

Figure 4: Covariate balance for DML and kappa estimators: Card specification, `educ16`

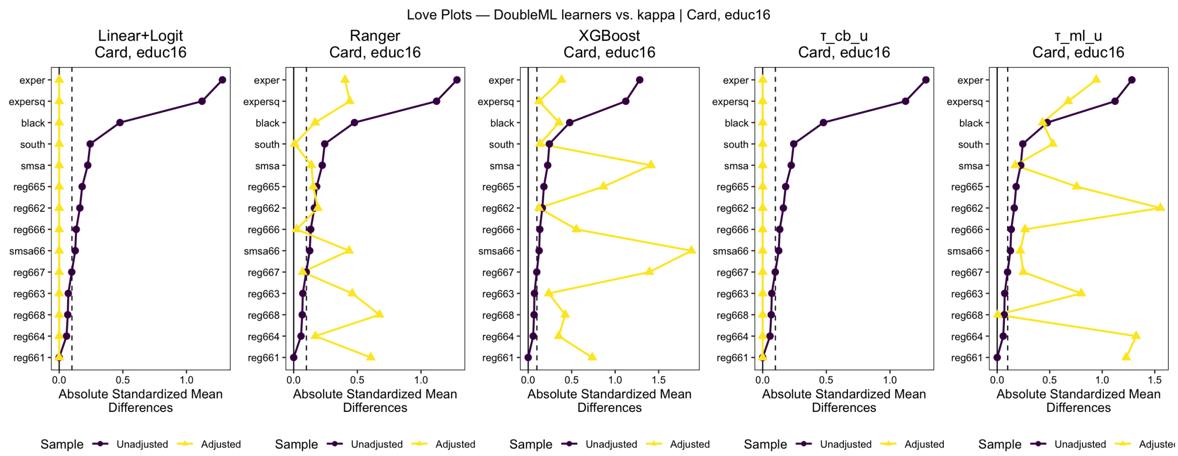

Figure 5: Learner sensitivity of Wald-AIPW covariate balance: Card specification, `educ16`

previous weight diagnostics, where XGBoost had very low modified effective sample size and extremely large maximum absolute weights. The Love plot therefore confirms that the XGBoost estimates are generated by an unstable weighted comparison rather than by a credible balanced contrast.

Those plots now add an additional layer to the analysis, showing whether the resulting weighted comparisons are credible in terms of observed covariates. Especially for the Card application, this depends on the estimator. CBPS-based normalized kappa produces the strongest covariate balance, linear/logit Wald-AIPW performs well, ranger is acceptable but less clean, and XGBoost fails both the weight-dispersion and balance diagnostics.

4 Empirical Child Application

A third central question in empirical labor economics is how child birth or the birth of an additional child affect mothers' labor-market outcomes. Families with more children often differ from families with fewer children in many observed and unobserved dimensions. Those can include preferences for work and family size, household resources and local labor-market conditions. A simple comparison between women with two children and women with more than two children would therefore not isolate the causal effect of childbearing. It would combine the effect of an additional child with selection into larger families. The third design based on the application of Angrist and Evans (1996) addresses this selection problem by using sibling sex composition as an instrument for fertility. This application complements the two previous empirical settings. The Vietnam draft lottery provides a low-dimensional, near-random-assignment benchmark, while the Card application studies an observational instrument with a richer covariate structure. The application based on Angrist and Evans (1996) instead combines a very large parent sample with binary labor-supply and log-income outcomes. It is therefore especially useful for the translation-invariance analysis, because the outcome coding of both labor-force participation and income is directly relevant for the behavior of normalized and unnormalized estimators. The next subsection introduces the data, the treatment and instrument definitions, and the two outcome variables used in the analysis.

4.1 Data and Design

The analysis follows Słoczyński et al. (2025) in revisiting the Angrist and Evans (1996) application based on the 1980 U.S. Census. The sample consists of women with at least two children. The empirical question is whether having an additional child affects mothers' labor-market outcomes. The treatment is therefore an indicator for whether a woman has more than two children,

$$D_i = \mathbf{1}\Big\{N_i^{\text{children}} > 2\Big\}.$$

The identifying variation comes from the sex composition of the first two children. Following Angrist and Evans (1996), the instrument is an indicator for whether the first two children are of the same sex,

$$Z_i = \mathbf{1}\{\text{boy}_{1i} = \text{boy}_{2i}\},$$

where boy_{1i} and boy_{2i} indicate whether the first and second child are boys. Equivalently, $Z_i = 1$ if the first two children are either two boys or two girls, and $Z_i = 0$ if the first two children are of mixed sex. The economic mechanism is that parents with two children of the same sex are more likely to have a third child, because some parents have a preference for a mixed-sex sibling composition. The instrument therefore shifts the probability of further childbearing but does not deterministically assign treatment. I consider two outcome variables. The first is labor-force participation, defined as an indicator for whether the woman worked for pay in the preceding year,

$$Y_i^{\text{LFP}} = \mathbf{1}\{\text{worked}_i = 1\}\,.$$

The second is log annual income,

$$Y_i^{\text{INC}} = \log(\text{income}_i)\,,$$

which is only defined for women with strictly positive reported income. These two outcomes are useful for different parts of the analysis. For the analysis, I use the following covariates, following the specification in Słoczyński et al. (2025), common to both outcome samples and all estimators. Table 19 lists them.

Table 19: Covariates used in the Angrist–Evans application

Covariate	Definition
age_i	Mother's age at census observation
agefirst_i	Mother's age at first birth
boy_{1i}	Indicator that the first child is a boy
boy_{2i}	Indicator that the second child is a boy
Black_i	Race indicator: Black
Hispanic_i	Ethnicity indicator: Hispanic
OtherRace_i	Race indicator: other race

4.1.1 Subsampling for the Angrist–Evans Application

Following Słoczyński et al. (2025), I use two corresponding parent samples from Farbmacher, Guber, and Vikström's 1980 Census extract. I use two corresponding parent samples. The first is the labor-force participation sample, defined for all 394,840 women with at least two children. The second is the log-income sample, restricted to the 220,502 women with strictly positive reported income. The full parent samples are too large for the computationally intensive diagnostics used below, especially for the estimation of the Smoother matrices. I therefore construct smaller diagnostic subsamples. Their purpose is to preserve the central empirical structure of the corresponding parent samples, in particular the instrument and treatment shares, the first-stage structure, and the relevant outcome distributions. To this end, I use proportional stratified subsampling without replacement. The procedure is applied separately to the labor-force participation and log-income parent samples, using a fixed seed. Each resulting diagnostic subsample contains 3,000 observations. The exact stratification variables and allocation rule are described in Appendix F.

5 Discussion

5.1 Cross-Application Summary

5.2 The Outcome-Weights Lens: What It Adds

5.3 DML Learner Comparison and Implications

5.4 Practical Guidance

5.5 Limitations

6 Conclusion

7 Appendix

A Outcome Weights of $\hat{\tau}_{a,1}$

This section derives the outcome weights of $\hat{\tau}_{a,1}$ (also written $\hat{\tau}_t$ in Słoczyński et al. 2025) in the PIVE framework of Knaus (2024), and establishes that it is untreated-unnormalized.

Setup. Define the raw instrument IPW contrast for observation i as

$$r_i = \frac{Z_i}{\hat{p}_i} - \frac{1-Z_i}{1-\hat{p}_i} = \frac{Z_i - \hat{p}_i}{\hat{p}_i(1-\hat{p}_i)}. \tag{36}$$

The two kappa components κ_{i1} and κ_{i0} decompose r_i as follows:

$$\kappa_{i1} = D_i \frac{Z_i - \hat{p}_i}{\hat{p}_i(1-\hat{p}_i)} = D_i r_i, \qquad \kappa_{i0} = (1-D_i)\frac{(1-Z_i)-(1-\hat{p}_i)}{\hat{p}_i(1-\hat{p}_i)} = -(1-D_i)r_i. \tag{37}$$

Equation (37) is the central identity: the treated part of r_i is κ_{i1} and the untreated part is $-\kappa_{i0}$.

PIVE representation. Define the diagonal transformation matrix $\mathbf{T}^{a,1} = \mathrm{diag}(r_1, \ldots, r_N)$. Using (37), the pseudo-treatment is $\mathbf{T}^{a,1}\mathbf{D} = \boldsymbol{\kappa}_1$, and the estimator can be written as

$$\hat{\tau}_{a,1} = \frac{N^{-1}\mathbf{1}_N'\mathbf{T}^{a,1}\mathbf{Y}}{N^{-1}\mathbf{1}_N'\mathbf{T}^{a,1}\mathbf{D}} = \frac{N^{-1}\sum_i r_i Y_i}{N^{-1}\sum_i \kappa_{i1}}, \tag{38}$$

which is a PIVE with $\widetilde{\mathbf{Z}} = \mathbf{1}_N$, $\widetilde{\mathbf{D}} = \boldsymbol{\kappa}_1$, $\widetilde{\mathbf{Z}}'\widetilde{\mathbf{D}} = \sum_i \kappa_{i1}$. By Proposition **??**, the outcome weights are

$$\boldsymbol{\omega}^{a,1\prime} = \left(\sum_i \kappa_{i1}\right)^{-1} \mathbf{1}_N'\mathbf{T}^{a,1}, \tag{39}$$

so that the weight of observation i is

$$\boxed{\omega_i^{a,1} = \frac{r_i}{\sum_j \kappa_{j1}} = \frac{1}{\sum_j \kappa_{j1}} \cdot \frac{Z_i - \hat{p}_i}{\hat{p}_i(1-\hat{p}_i)}.} \tag{40}$$

Normalization properties. *(i) Total weight sum.* Let $s_1 = \sum_i Z_i/\hat{p}_i$ and $s_0 = \sum_i (1-Z_i)/(1-\hat{p}_i)$. Then $\sum_i r_i = s_1 - s_0$, so

$$\sum_{i=1}^N \omega_i^{a,1} = \frac{\sum_i r_i}{\sum_j \kappa_{j1}} = \frac{s_1 - s_0}{\sum_j \kappa_{j1}}, \tag{41}$$

which is zero only if $s_1 = s_0$, a coincidental data-dependent condition with no structural guarantee. Hence $\sum_i \omega_i^{a,1} \neq 0$ in general, and $\hat{\tau}_{a,1}$ is not translation invariant.

(ii) Treated weight sum. Using $D_i r_i = \kappa_{i1}$ from (37):

$$\sum_{i=1}^{N} \omega_i^{a,1} D_i = \frac{\sum_i D_i r_i}{\sum_j \kappa_{j1}} = \frac{\sum_i \kappa_{i1}}{\sum_j \kappa_{j1}} = 1. \qquad \checkmark \tag{42}$$

The treated weights sum to one exactly, because the denominator of $\hat{\tau}_{a,1}$ is $\sum_i \kappa_{i1}$ — precisely the treated-side kappa total.

(iii) Untreated weight sum. Using $(1 - D_i)r_i = -\kappa_{i0}$ from (37):

$$\sum_{i=1}^{N} \omega_i^{a,1}(1 - D_i) = \frac{\sum_i (1 - D_i) r_i}{\sum_j \kappa_{j1}} = \frac{-\sum_i \kappa_{i0}}{\sum_j \kappa_{j1}}, \tag{43}$$

which equals -1 only if $\sum_i \kappa_{i0} = \sum_i \kappa_{i1}$. Since there is no structural reason for the complier shares from the treated and untreated sides to be equal in any given sample, $\sum_i \omega_i^{a,1}(1 - D_i) \neq -1$ in general.

Classification. Collecting the three results:

$$\sum_{i=1}^{N} \omega_i^{a,1} \neq 0, \qquad \sum_{i=1}^{N} \omega_i^{a,1} D_i = 1, \qquad \sum_{i=1}^{N} \omega_i^{a,1}(1 - D_i) = -\frac{\sum_i \kappa_{i0}}{\sum_i \kappa_{i1}} \neq -1. \tag{44}$$

Therefore $\hat{\tau}_{a,1}$ is *untreated-unnormalized* in the sense of Knaus (2024, Table 4): the treated side is normalized by construction but the untreated side is not. The contrast with $\hat{\tau}_{a,10}$ is instructive: the normalized estimator uses separate denominators $\sum_i \kappa_{i1}$ and $\sum_i \kappa_{i0}$ for the two sides, forcing both to sum to their target values; $\hat{\tau}_{a,1}$ uses only the treated-side denominator, leaving the untreated side without its own rescaling.

B R Code Overview

The empirical analysis relies on a created set of purpose-built R functions that implement the outcome-weight framework derived in Section ??. The functions are currently contained in `functions_kappa.R`; the final replication package, whether as a standalone script, a collection of application-specific files, or an R package will be made available at the link at the end of this appendix.

The central contribution of the code is the function `kappa_outcome_weights(Z, D, p)`, which takes the observed instrument Z, treatment D, and a fitted propensity score vector $\hat{p}$ and returns the five outcome weight vectors $\boldsymbol{\omega}^{u}$, $\boldsymbol{\omega}^{a,10}$, $\boldsymbol{\omega}^{a}$, $\boldsymbol{\omega}^{a,1}$, and $\boldsymbol{\omega}^{a,0}$ in closed form. The weights are computed directly from the expressions derived in Section ??, without numerical optimisation or outcome regression. For $\hat{\tau}_u$, the function applies the Hájek-normalized IPW contrast from (??); for $\hat{\tau}_{a,10}$, it applies the separately normalised κ_1/κ_0 ratio; for the three unnormalized estimators, it uses the common IPW numerator with the three denominator choices from Słoczyński et al. (2025).

Listing 1: `kappa_outcome_weights()`: closed-form outcome weights for all five kappa estimators

```r
kappa_outcome_weights <- function(Z, D, p) {
  kw  <- kappa_weights(Z, D, p)          # kappa, kappa1, kappa0

  # tau_u: Hajek-normalized IPW contrast (eq. u_omega_scalar)
  s1  <- sum(Z / p);  s0 <- sum((1 - Z) / (1 - p))
  dD  <- sum(D * Z / p) / s1 - sum(D * (1 - Z) / (1 - p)) / s0
  w_u <- (Z / p / s1 - (1 - Z) / (1 - p) / s0) / dD

  # tau_a10: separately normalized kappa1 / kappa0
  w_a10 <- kw$kappa1 / sum(kw$kappa1) - kw$kappa0 / sum(kw$kappa0)

  # tau_a, tau_a1, tau_a0: common IPW numerator, three denominators
  num_w <- (Z - p) / (p * (1 - p)) / length(Z)

  list(w_u   = as.vector(w_u),
       w_a10 = as.vector(w_a10),
       w_a   = as.vector(num_w / mean(kw$kappa)),
       w_a1  = as.vector(num_w / mean(kw$kappa1)),
       w_a0  = as.vector(num_w / mean(kw$kappa0)))
```

```
}
```

Two companion functions produce the diagnostics reported in every weight table throughout Chapters 2 and 3. `weight_diag(w)` returns the four quantities that appear in each row of Tables 7 and **??**: the weight sum $\sum_i \omega_i$ (the translation-invariance criterion of (**??**)), the effective sample size $\mathrm{ESS} = 1/\sum_i \omega_i^2$, the share of negative weights, and the maximum absolute weight. `check_weight_identity(w, Y, tau_hat)` verifies the identity $\hat{\tau} = \sum_i \omega_i Y_i$ to machine precision for every point estimate reported; all checks pass across all specifications. The propensity score is estimated either by standard logit MLE via `logit_mle()` or by a hand-coded CBPS solver, `fit_cbps_alpha()`, which implements Newton steps with backtracking line search to solve the covariate-balancing moment condition directly. Analytical standard errors follow from `kappa_analytic_se_one()`, which stacks the propensity-score and LATE moment conditions and computes the M-estimation sandwich via a numerical Jacobian.

All replication code is available at: `[GITHUB / CLOUD LINK]`

C Instrument Propensity Scores

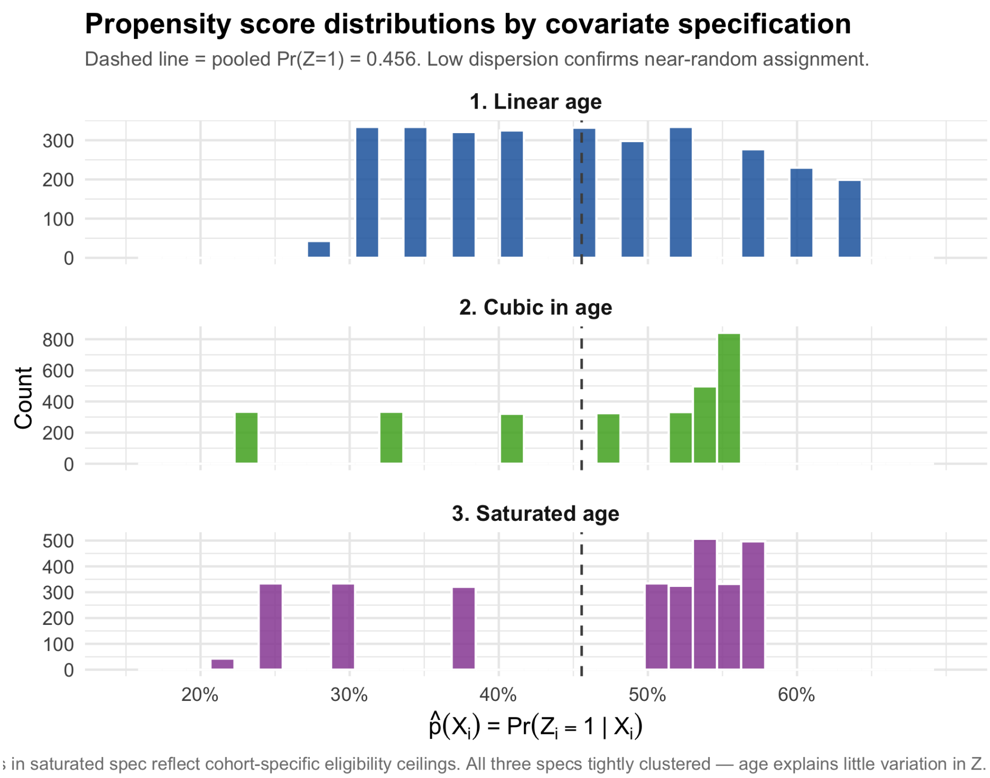

Figure 6: The first Empirical Application : Distribution of the fitted propensity score $\hat{p}(X_i) = \Pr(Z_i = 1 \mid X_i)$ under three covariate specifications.

D SUW replication

Table 20: Kappa estimators: log wages (Vietnam draft lottery)

	Linear age		Cubic age		Saturated age	
Estimator	Cents	Dollars	Cents	Dollars	Cents	Dollars
Panel A: 2SLS benchmark						
2SLS	0.233	0.233	0.227	0.227	0.254	0.254
	(0.212)	(0.212)	(0.229)	(0.229)	(0.227)	(0.227)
Panel B: Normalised kappa estimators						
$\hat{\tau}_u^{cb}$	0.229	0.229	0.210	0.210	0.241	0.241
	(0.213)	(0.213)	(0.233)	(0.233)	(0.229)	(0.229)
$\hat{\tau}_u^{ml}$	0.234	0.234	0.202	0.202	0.241	0.241
	(0.211)	(0.211)	(0.235)	(0.235)	(0.229)	(0.229)
$\hat{\tau}_{a,10}$	0.227	0.227	0.204	0.204	0.241	0.241
	(0.204)	(0.204)	(0.239)	(0.239)	(0.229)	(0.229)
Panel C: Unnormalised kappa estimators						
$\hat{\tau}_a$	-0.429^{*}	0.015	0.537^{*}	0.314	0.241	0.241
	(0.258)	(0.207)	(0.322)	(0.252)	(0.229)	(0.229)
$\hat{\tau}_t$	-0.455	0.016	0.515^{*}	0.302	0.241	0.241
	(0.279)	(0.219)	(0.301)	(0.240)	(0.229)	(0.229)
$\hat{\tau}_{a,0}$	-0.413^{*}	0.014	0.540^{*}	0.317	0.241	0.241
	(0.246)	(0.199)	(0.326)	(0.255)	(0.229)	(0.229)

The two outcomes differ by $\log(100) \approx 4.605$. The cents/dollars comparison is a finite-sample test of translation invariance: an estimator is translation invariant if and only if $\sum_i \omega_i = 0$, which holds for all estimators in Panels A and B and fails for those in Panel C. Covariate specifications: linear age (columns 1–2), cubic polynomial in age (columns 3–4), and age indicators (columns 5–6). The propensity score $\hat{p}(X_i) = \Pr(Z_i = 1 \mid X_i)$ is estimated by logistic MLE except for $\hat{\tau}_u^{cb}$, which uses CBPS moment conditions. Standard errors: HC1-robust for 2SLS; M-estimation sandwich for kappa estimators. $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$. Replicates Słoczyński, Uysal, & Wooldridge (2025), Table 2.

E DML learner comparison

Table 21: DoubleML learner comparison: Wald-AIPW (Vietnam, cubic age)

Learner	Est.	SE	$\omega'\mathbf{Y} = \hat{\tau}$	$\sum_i \omega_i$
Linear + logistic	0.2178	(0.233)	✓	0.000
Ranger RF	0.2461	(0.234)	✓	0.000
XGBoost†	0.2463	(0.234)	×	-3.7×10^{-6}

Notes: All models use `DoubleMLIIVM` with five-fold cross-fitting (**?**)

F Subsampling Procedure for the Card framework

The diagnostic subsamples used in the Angrist–Evans application are constructed by proportional stratified subsampling without replacement. Let Z_i denote the same-sex instrument and D_i the more-than-two-children treatment. For the labor-force participation sample, the strata are defined by Z_i, D_i, the binary labor-force outcome, positive-income status, race, and mother's age group. For the log-income sample, the strata are defined by Z_i, D_i, income decile, race, and mother's age group. For each parent sample, let N_0 denote the parent-sample size and let n_s denote the size of stratum $s = 1, \ldots, S$. The target subsample size is set to $N = 3{,}000$. The proportional target allocation for each stratum is

$$n_s^* = N \frac{n_s}{N_0}.$$

The preliminary allocation is obtained by taking the integer part, $\lfloor n_s^* \rfloor$. The remaining observations,

$$N - \sum_{s=1}^{S} \lfloor n_s^* \rfloor,$$

are then assigned one at a time to the strata with the largest fractional remainders, until the allocated total equals N exactly. Within each stratum, the allocated number of observations is drawn randomly without replacement. The procedure is implemented separately for the labor-force participation and log-income parent samples, using a fixed seed. This yields two diagnostic subsamples of 3,000 observations each.

References

Abadie, A. (2003). Semiparametric instrumental variable estimation of treatment response models. *Journal of econometrics*, 113(2):231–263.

Angrist, J. and Evans, W. N. (1996). Children and their parents' labor supply: Evidence from exogenous variation in family size.

Angrist, J. D. (1990). Lifetime earnings and the vietnam era draft lottery: evidence from social security administrative records. *The american economic review*, pages 313–336.

Angrist, J. D., Imbens, G. W., and Rubin, D. B. (1996). Identification of causal effects using instrumental variables. *Journal of the American statistical Association*, 91(434):444–455.

Angrist, J. D. and Krueger, A. B. (1991). Does compulsory school attendance affect schooling and earnings? *The quarterly journal of economics*, 106(4):979–1014.

Blandhol, C., Bonney, J., Mogstad, M., and Torgovitsky, A. (2022). When is tsls actually late? Technical report, National Bureau of Economic Research Cambridge, MA.

Card, D. (1993). Using geographic variation in college proximity to estimate the return to schooling.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters.

Griliches, Z. (1977). Estimating the returns to schooling: Some econometric problems. *Econometrica: Journal of the Econometric Society*, pages 1–22.

Kitagawa, T. (2015). A test for instrument validity. *Econometrica*, 83(5):2043–2063.

Knaus, M. C. (2024). Treatment effect estimators as weighted outcomes. *arXiv preprint arXiv:2411.11559*.

Słoczyński, T., Uysal, S. D., and Wooldridge, J. M. (2025). Abadie's kappa and weighting estimators of the local average treatment effect. *Journal of Business & Economic Statistics*, 43(1):164–177.

Wald, A. (1940). The fitting of straight lines if both variables are subject to error. *The annals of mathematical statistics*, 11(3):284–300.

Wooldridge, J. M. (2016). *Introductory econometrics a modern approach*. South-Western cengage learning.